Chemistry: CYL1010

Textbooks

1. Silberberg, M., *Chemistry: The Molecular Nature of Matter and Change*, 6^{th} Edition, McGraw Hill Education.
2. West, A.R., (2015), *Solid State Chemistry and Its Applications*, 2^{nd} edition, John Wiley & Sons.
3. J. D. Lee, Concise Inorganic Chemistry, 5^{th} Edition.
4. Atkins, Physical Chemistry, 11^{th} Edition.

Reference Books

1. McMurry, J. E. & Fay, R. C. *Chemistry*, 5^{th} Edition, Pearson.
2. Jonathan Clayden, Nick Greeves, Stuart Warren, Organic Chemistry, 2nd Edition.
3. F. Albert Cotton, Geoffrey Wilkinson, Basic Inorganic Chemistry, Wiley Student Edition.

Self Learning Material

1. https://nptel.ac.in/downloads/122101001/

Transition Metal Chemistry [5 Lectures]: Colors in Complexes, Organometallics and Catalysis, Bioinorganic Chemistry

Organic Chemistry [8 Lectures]: Name Reactions, Stereochemistry and their Applications, Acid-Base Chemistry, Use of Buffers, Organic Chemistry examples such as polymers and biopolymers, drugs and photoactive molecules in everyday life and advanced technologies

Atomic number → 78
Symbol → Pt
Name → Platinum
Atomic mass → 195.08
Electron per shell → 2-8-18-32-17-1

Alkali metals	Alkaline earth metals	Transition metals
Lanthanides	Actinides	Post transition metals
Metalloids	Non-metals	Noble gases

s-block | d-block | p-block

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1 H Hydrogen 1.008 1																	2 He Helium 4.0026 2
3 Li Lithium 6.940 2-1	4 Be Beryllium 9.012 2-1											5 B Boron 10.810 2-3	6 C Carbon 12.011 2-4	7 N Nitrogen 14.007 2-5	8 O Oxygen 15.999 2-6	9 F Fluorine 18.998 2-7	10 Ne Neon 20.180 2-8
11 Na Sodium 22.99 2-8-1	12 Mg Magnesium 24.30 2-8-2											13 Al Aluminium 26.98 2-8-3	14 Si Silicon 28.085 2-8-4	15 P Phosphorus 30.97 2-8-5	16 S Sulfur 32.06 2-8-6	17 Cl Chlorine 35.45 2-8-7	18 Ar Argon 39.95 2-8-8
19 K Potassium 39.098 2-8-8-1	20 Ca Calcium 40.078 2-8-8-2	21 Sc Scandium 44.956 2-8-9-2	22 Ti Titanium 47.867 2-8-10-2	23 V Vanadium 50.942 2-8-11-2	24 Cr Chromium 51.996 2-8-13-1	25 Mn Manganese 54.938 2-8-13-2	26 Fe Iron 55.845 2-8-14-2	27 Co Cobalt 58.933 2-8-15-2	28 Ni Nickel 58.693 2-8-16-2	29 Cu Copper 63.546 2-8-18-1	30 Zn Zinc 63.380 2-8-18-2	31 Ga Gallium 69.723 2-8-18-3	32 Ge Germanium 72.630 2-8-18-4	33 As Arsenic 74.922 2-8-18-5	34 Se Selenium 78.971 2-8-18-6	35 Br Bromine 79.904 2-8-18-7	36 Kr Krypton 83.798 2-8-18-8
37 Rb Rubidium 85.468 2-8-18-8-1	38 Sr Strontium 87.620 2-8-18-8-2	39 Y Yttrium 88.906 2-8-18-9-2	40 Zr Zirconium 91.224 2-8-18-10-2	41 Nb Niobium 92.906 2-8-18-12-1	42 Mo Molybdenum 95.950 2-8-18-13-1	43 Tc Technetium [97] 2-8-18-13-2	44 Ru Ruthenium 101.07 2-8-18-15-1	45 Rh Rhodium 102.91 2-8-18-16-1	46 Pd Palladium 106.42 2-8-18-18	47 Ag Silver 107.87 2-8-18-18-1	48 Cd Cadmium 112.41 2-8-18-18-2	49 In Indium 114.82 2-8-18-18-3	50 Sn Tin 118.71 2-8-18-18-4	51 Sb Antimony 121.76 2-8-18-18-5	52 Te Tellurium 127.60 2-8-18-18-6	53 I Iodine 126.90 2-8-18-18-7	54 Xe Xenon 131.29 2-8-18-18-8
55 Cs Cesium 132.91 2-8-18-18-8-1	56 Ba Barium 137.33 2-8-18-18-8-2	57 - 71 Ln Lanthanides	72 Hf Hafnium 178.49 2-8-18-32-10-2	73 Ta Tantalum 180.95 2-8-18-32-11-2	74 W Tungsten 183.84 2-8-18-32-12-2	75 Re Rhenium 186.21 2-8-18-32-13-2	76 Os Osmium 190.23 2-8-18-32-14-2	77 Ir Iridium 192.22 2-8-18-32-15-2	78 Pt Platinum 195.08 2-8-18-32-17-1	79 Au Gold 196.97 2-8-18-32-18-1	80 Hg Mercury 200.59 2-8-18-32-18-2	81 Tl Thallium 204.38 2-8-18-32-18-3	82 Pb Lead 207.20 2-8-18-32-18-4	83 Bi Bismuth 208.98 2-8-18-32-18-5	84 Po Polonium [209] 2-8-18-32-18-6	85 At Astatine [210] 2-8-18-32-18-7	86 Rn Radon [222] 2-8-18-32-18-8
87 Fr Francium [223] 2-8-18-32-18-8-1	88 Ra Radium [226] 2-8-18-32-18-8-2	89 - 103 Actinides	104 Rf Rutherfordium [267] 2-8-18-32-32-10-2	105 Db Dubnium [268] 2-8-18-32-32-11-2	106 Sg Seaborgium [269] 2-8-18-32-32-12-2	107 Bh Bohrium [270] 2-8-18-32-32-13-2	108 Hs Hassium [277] 2-8-18-32-32-14-2	109 Mt Meitnerium [278] 2-8-18-32-32-15-2	110 Ds Darmstadtium [281] 2-8-18-32-32-16-2	111 Rg Roentgenium [282] 2-8-18-32-32-17-2	112 Cn Copernicium [285] 2-8-18-32-32-18-2	113 Nh Nihonium [286] 2-8-18-32-32-18-3	114 Fl Flerovium [289] 2-8-18-32-32-18-4	115 Mc Moscovium [290] 2-8-18-32-32-18-5	116 Lv Livermorium [293] 2-8-18-32-32-18-6	117 Ts Tennessine [294] 2-8-18-32-32-18-7	118 Og Oganesson [294] 2-8-18-32-32-18-8

f-block

57 La Lanthanum 138.91 2-8-18-18-9-2	58 Ce Cerium 140.12 2-8-18-19-9-2	59 Pr Praseodymium 140.91 2-8-18-21-8-2	60 Nd Neodymium 144.24 2-8-18-22-8-2	61 Pm Promethium [145] 2-8-18-23-8-2	62 Sm Samarium 150.36 2-8-18-24-8-2	63 Eu Europium 151.96 2-8-18-25-8-2	64 Gd Gadolinium 157.25 2-8-18-25-9-2	65 Tb Terbium 158.93 2-8-18-27-8-2	66 Dy Dysprosium 162.50 2-8-18-28-8-2	67 Ho Holmium 164.93 2-8-18-29-8-2	68 Er Erbium 167.26 2-8-18-30-8-2	69 Tm Thulium 168.93 2-8-18-31-8-2	70 Yb Ytterbium 173.05 2-8-18-32-8-2	71 Lu Lutetium 174.97 2-8-18-32-9-2
89 Ac Actinium [227] 2-8-18-32-18-9-2	90 Th Thorium 232.04 2-8-18-32-18-10-2	91 Pa Protactinium 231.04 2-8-18-32-20-9-2	92 U Uranium 238.03 2-8-18-32-21-9-2	93 Np Neptunium [237] 2-8-18-32-22-9-2	94 Pu Plutonium [244] 2-8-18-32-24-8-2	95 Am Americium [243] 2-8-18-32-25-8-2	96 Cm Curium [247] 2-8-18-32-25-9-2	97 Bk Berkelium [247] 2-8-18-32-27-8-2	98 Cf Californium [251] 2-8-18-32-28-8-2	99 Es Einsteinium [252] 2-8-18-32-29-8-2	100 Fm Fermium [257] 2-8-18-32-30-8-2	101 Md Mendelevium [258] 2-8-18-32-31-8-2	102 No Nobelium [259] 2-8-18-32-32-8-2	103 Lr Lawrencium [266] 2-8-18-32-32-8-3

Bonding in Transition Metal Complexes

- **Werner Coordination Theory**
- **Valence Bond Theory**
- **Crystal Field Theory**

Werner Coordination Theory

- Compounds made up of simpler compounds are called coordination compounds.
- $CoCl_3$ and NH_3.
  - $CoCl_3 \cdot (NH_3)_6$ and $CoCl_3 \cdot (NH_3)_5$.
  - Differing reactivity with $AgNO_3$.

Alfred Werner

Nobel Prize in 1913

Werner Coordination Theory

Cobalt (III) Complexes: $CoCl_2 + NH_3$ (aq)

$CoCl_3$ 6 NH_3	Orange-yellow	$CoCl_3$ 5 NH_3 H_2O	Red
$CoCl_3$ 5 NH_3	Purple	$CoCl_3$ 4 NH_3	Green

$$NH_{3\ (aq)} + HCl_{\ (aq)} \longrightarrow NH_4^+{}_{(aq)} + Cl^-{}_{(aq)}$$

$$CoCl_3\ 6NH_{3\ (aq)} + HCl_{\ (aq)} \longrightarrow \text{No Reaction}$$

$$Ag^+{}_{(aq)} + Cl^-{}_{(aq)} \longrightarrow AgCl\ (s)$$

Expected	**Observed**
$CoCl_3\ 6NH_3 + Ag^+ \longrightarrow 3\ AgCl\ (s)$	$CoCl_3\ 6NH_3 + Ag^+ \longrightarrow 3\ AgCl\ (s)$
$CoCl_3\ 5NH_3 + Ag^+ \longrightarrow 3\ AgCl\ (s)$	$CoCl_3\ 6NH_3 + Ag^+ \longrightarrow 2\ AgCl\ (s)$
$CoCl_3\ 4NH_3 + Ag^+ \longrightarrow 3\ AgCl\ (s)$	$CoCl_3\ 6NH_3 + Ag^+ \longrightarrow 1\ AgCl\ (s)$

Conductivity Measurements

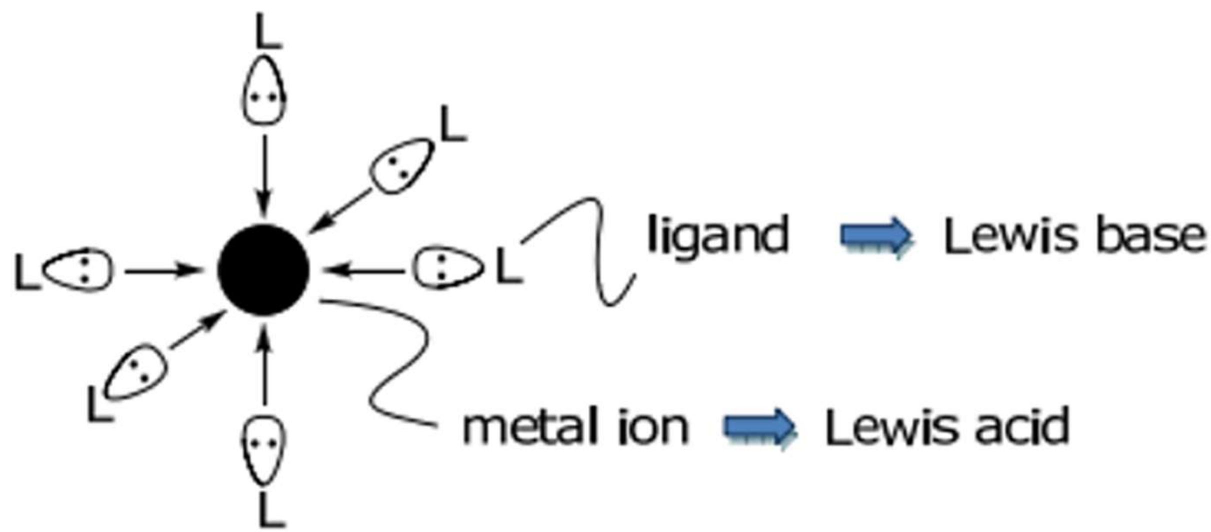
$CoCl_3\ 6\ NH_3$	Four Ions	$CoCl_3\ 5\ NH_3\ H_2O$	Four Ions
$CoCl_3\ 5\ NH_3$	Three Ions	$CoCl_3\ 4\ NH_3$	Two Ions

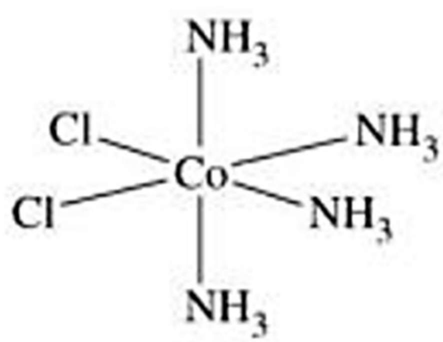
Two types of valence or bonding capacity

- Primary valence: Based on the number of electrons an atom loses in forming the ion. Also called as ionizable valence and corresponds to oxidation state.
- Secondary valence: Responsible for the bonding of other groups, called ligands, to the central metal atom. Werner assumed that the secondary valence of the transition metal in these Cobalt (III) complexes is six.

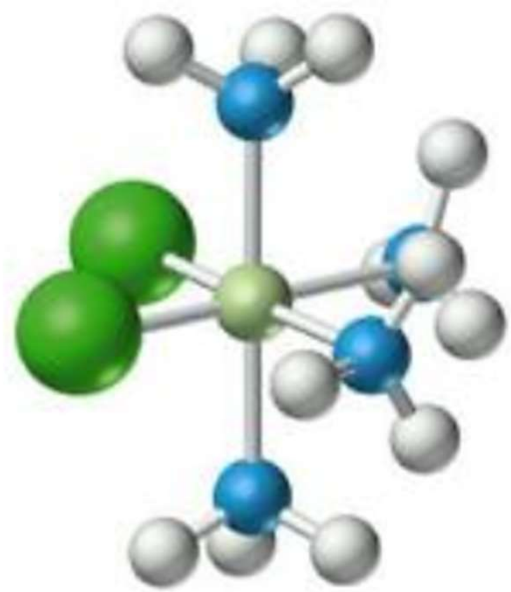
$[Co(NH_3)_6]^{3+}\ 3Cl^-$	Orange-yellow	$[Co(NH_3)_5Cl]^{2+}\ 2Cl^-$	Purple
$[Co(NH_3)_5(H_2O)]^{3+}\ 3Cl^-$	Red	$[Co(NH_3)_4Cl_2]^+\ Cl^-$	Green

Metal complexes are Lewis acid-base adducts formed between metal ions (the acid) and ligands (the base).





cis-$[CoCl_2(NH_3)_4]^+$
(purple)



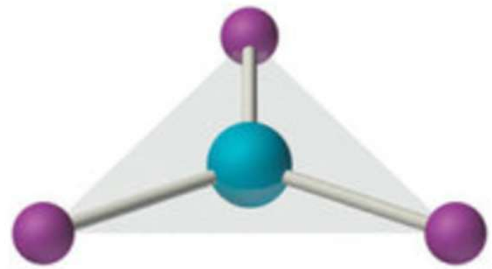
Coordination Number

2

Linear ML_2

$[Au(CN)_2]^-$

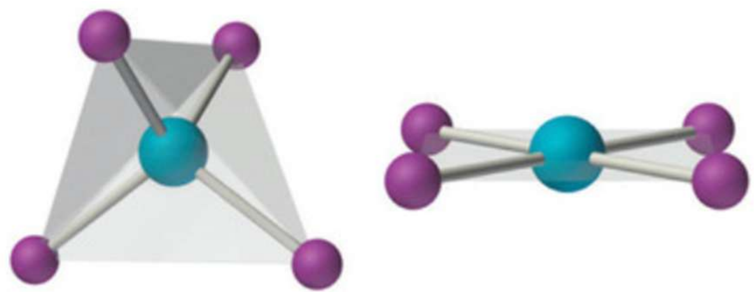
3



Trigonal Planar ML_3

$[HgI_3]^-$

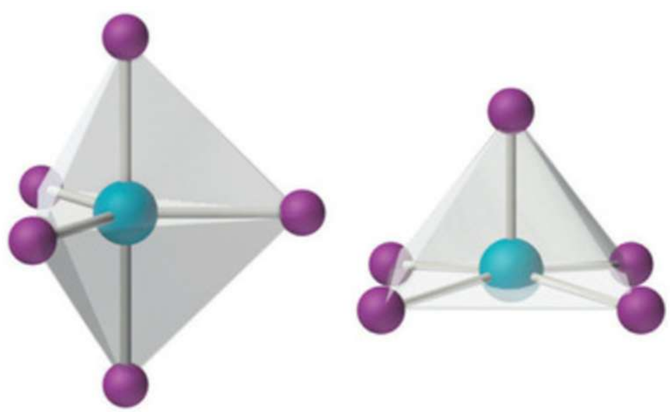
4



Tetrahedral and Square Planar ML_4

$[FeCl_4]^-$

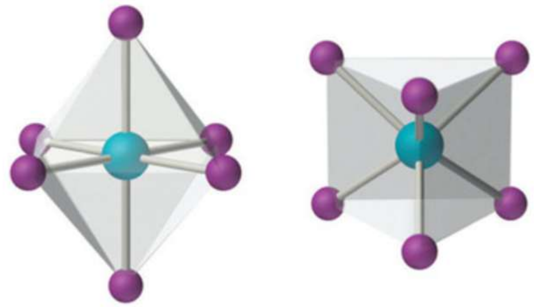
5



Trigonal Bipyramidal and Square Pyramidal ML_5

$[Cu(Cl_5)]^{3-}$

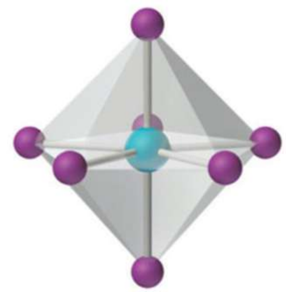
6



Octahedral and Trigonal Prismatic ML_6

$[Co(NH_3)_6]^{3+}$

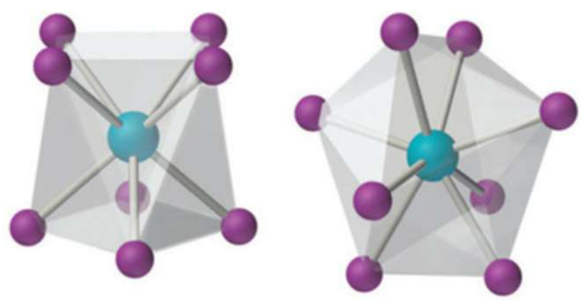
7



Pentagonal Bipyramidal ML_7

$[ZrF_7]^{3-}$

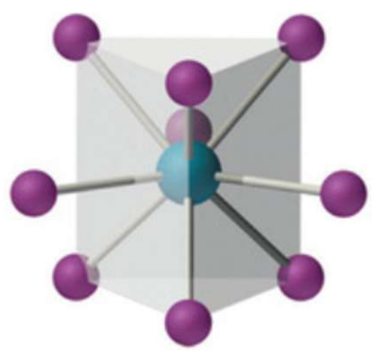
8



Square Antiprismatic and Trigonal Dodecahedral ML_8

$[Mo(CN)_8]^{4-}$

9



Tricapped Trigonal Prismatic ML_9

$[Nd(H_2O)_9]^{3+}$

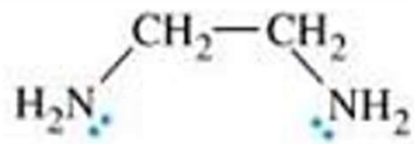
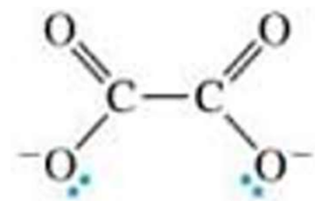
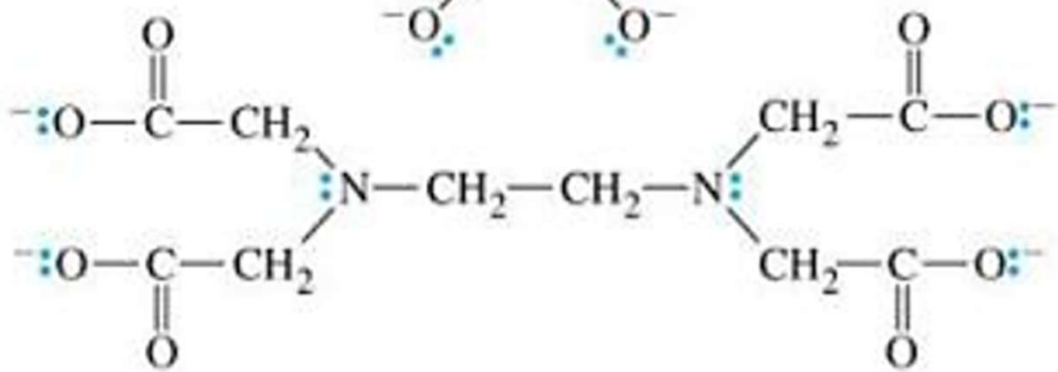
Ligands

Formula	Name as Ligand	Formula	Name as Ligand	Formula	Name as Ligand
Neutral molecules		**Anions**		**Anions**	
H_2O	Aqua	F^-	Fluoro	SO_4^{2-}	Sulfato
NH_3	Ammine	Cl^-	Chloro	$S_2O_3^{2-}$	Thiosulfato
CO	Carbonyl	Br^-	Bromo	NO_2^-	Nitrito-*N*-[a]
NO	Nitrosyl	I^-	Iodo	ONO^-	Nitrito-*O*-[a]
CH_3NH_2	Methylamine	O^{2-}	Oxo	SCN^-	Thiocyanato-*S*-[b]
C_5H_5N	Pyridine	OH^-	Hydroxo	NCS^-	Thiocyanato-*N*-[b]
		CN^-	Cyano		

[a] If the nitrite ion is attached through the N atom ($-NO_2$), the designation *nitrito-N-* is used; if attached through an O atom (—ONO), *nitrito-O-*.

[b] If the thiocyanate ion is attached through the S atom (—SCN), the name *thiocyanato -S-* is used; if attachment is through the N atom (—NCS), *thiocyanato-N-*.

Polydentate Ligands

Abbreviation	Name	Formula
en	Ethylenediamine	
ox^{2-}	Oxalato	
$EDTA^{4-}$	Ethylenediaminetetraacetato	

Valence Bond Theory

G. N. Lewis (1902): Atoms form covalent bonds by sharing electron pair.

W. Heitler and F. London (1927): Showed how the sharing of pairs of electrons holds a covalent molecule together. The Heitler-London model of covalent bonds was the basis of the VBT.

L. Pauling: Atomic orbitals are mixed to form hybrid orbitals such as sp, sp^2, sp^3, dsp^2, dsp^3, and d^2sp^3 orbitals.

Valence Bond Theory: Assumptions/Features

- Ligands form covalent-coordinate bonds to the metal.
- Ligands must have lone pair of electrons.
- Available empty orbital of suitable energy for metal for bonding.
- Atomic (hybrid) orbitals are used for bonding (rather than molecular orbitals)

Can explain: Shape and stability of the metal complex.

Cannot explain: Color and temperature dependence of magnetic properties

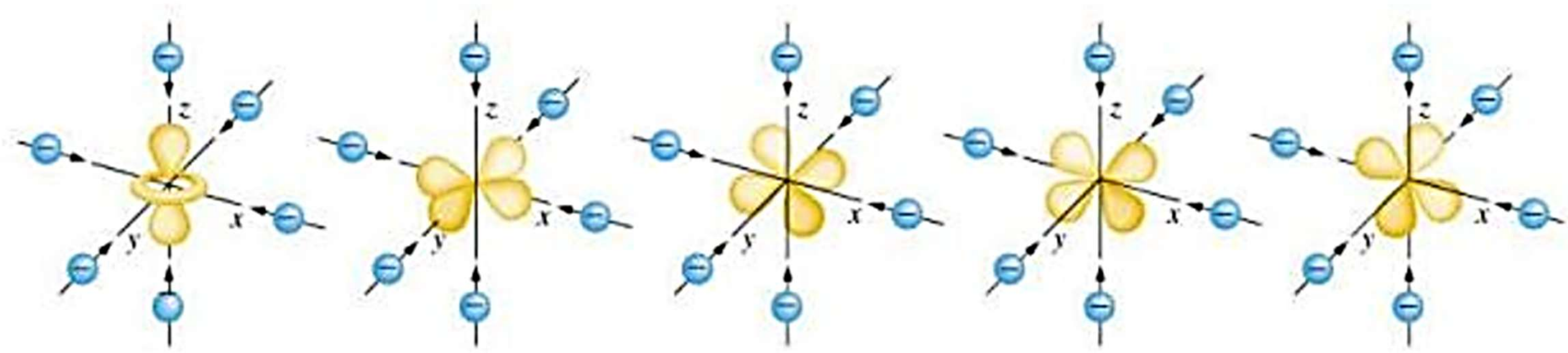
Distortion in Metal Complexes

Crystal Field Theory

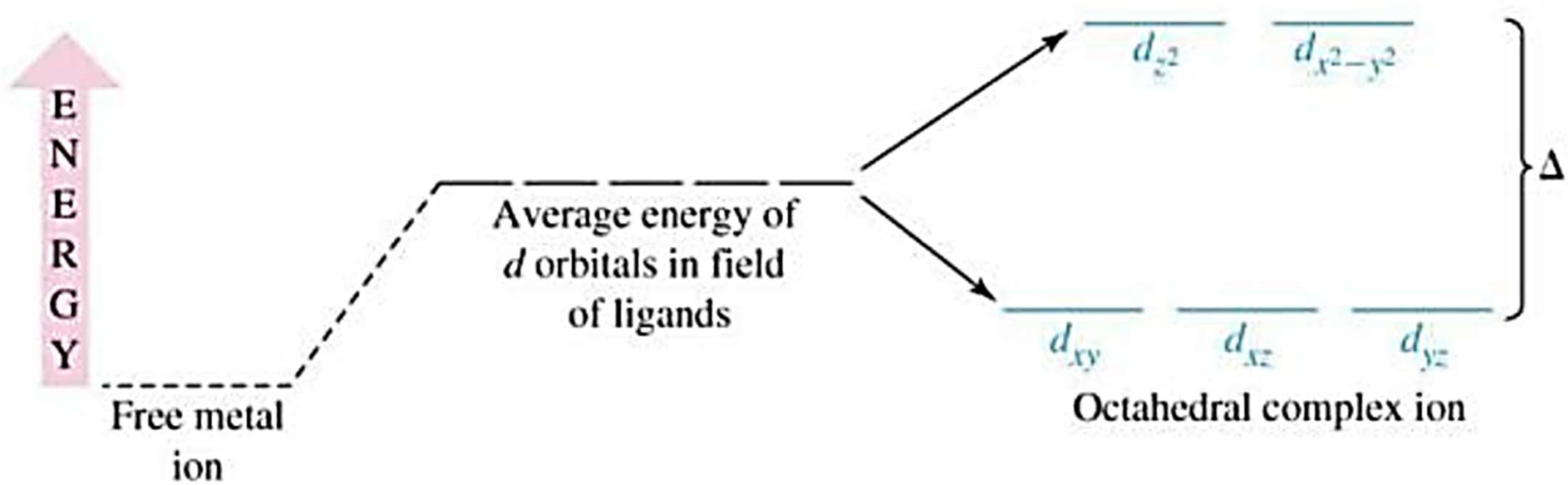
Consider bonding in a complex to be an electrostatic attraction between a positively charged nucleus and the electrons of the ligands.

- Electrons on metal atom repel electrons on ligands.
- Focus particularly on the d-electrons on the metal ion.
- Negative ligand: ion-ion interaction, neutral ligand: ion-dipole interaction
- The electrons on the metal occupy those d-orbitals farthest away from the direction of approach of ligands.

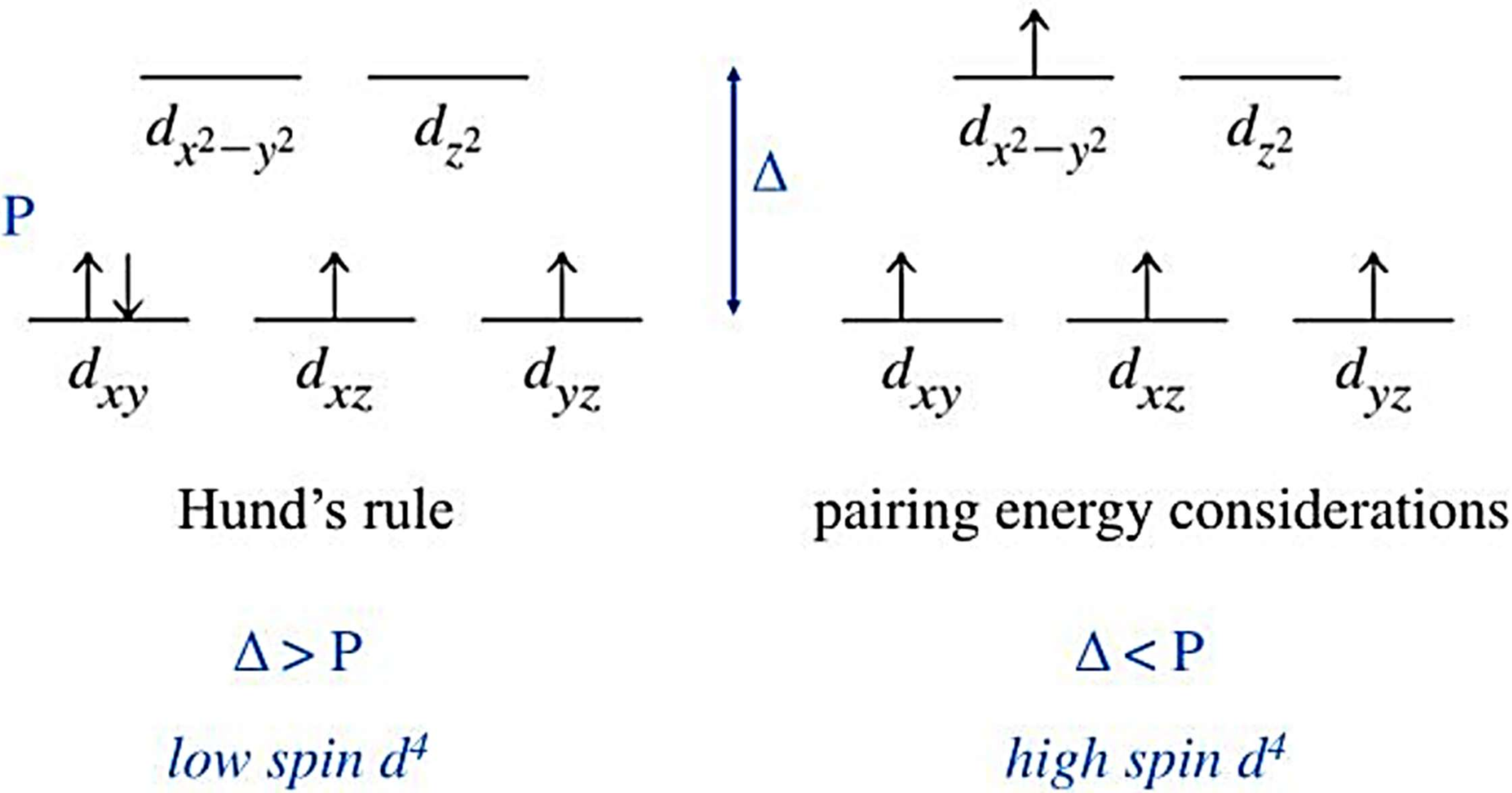
Octahedral Complex and *d*-orbital Energies



(a) d_{z^2} (b) $d_{x^2-y^2}$ (c) d_{xy} (d) d_{xz} (e) d_{yz}



Electronic Configuration in *d*-orbital



For d^1-d^3 systems: t_{2g}^3 (Hund's rule of maximum multiplicity)

For d^4-d^7 systems:

$t_{2g}^4 e_g^0$ (low spin case or strong field situation)

$t_{2g}^3 e_g^1$ (high spin case or weak field situation)

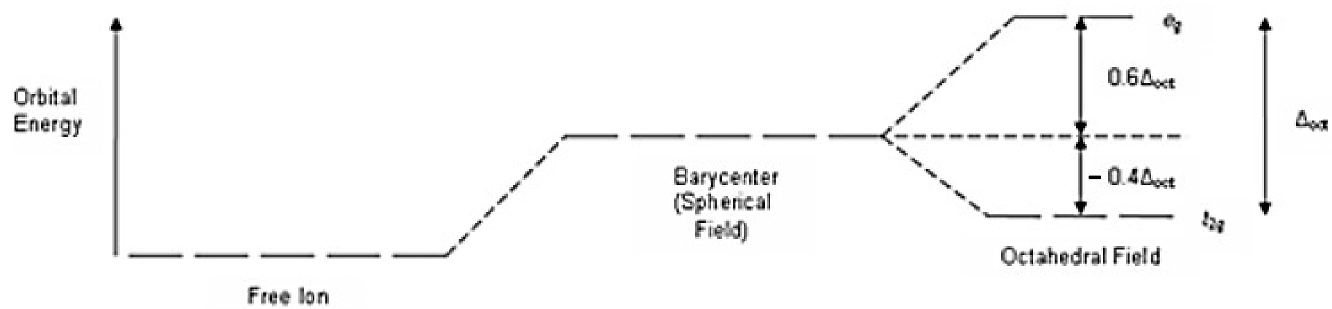
Parameters (for HS and LS case):

CFSE (value of Δ_o)

Pairing energy of e^- (repulsive)

CFSE > P.E.: LS Complex

CFSE < P.E.: HS Complex



Conservation of barycenter from a spherical field to octahedral field indicates t_{2g} set must be stabilized as much as the e_g set is destabilized.

$[CrCl_6]^{3-}$	13640 cm^{-1}
$[Cr(H_2O)_6]^{3+}$	17830
$[Cr(NH_3)_6]^{3+}$	21680
$[Cr(CN)_6]^{3-}$	26280
$[Co(NH_3)_6]^{3+}$	24800
$[Rh(NH_3)_6]^{3+}$	34000
$[Ir(NH_3)_6]^{3+}$	41000

3d < 4d < 5d

$M^{2+} < M^{3+} < M^{4+}$

Spectrochemical Series

Small Δ
Weak field ligands

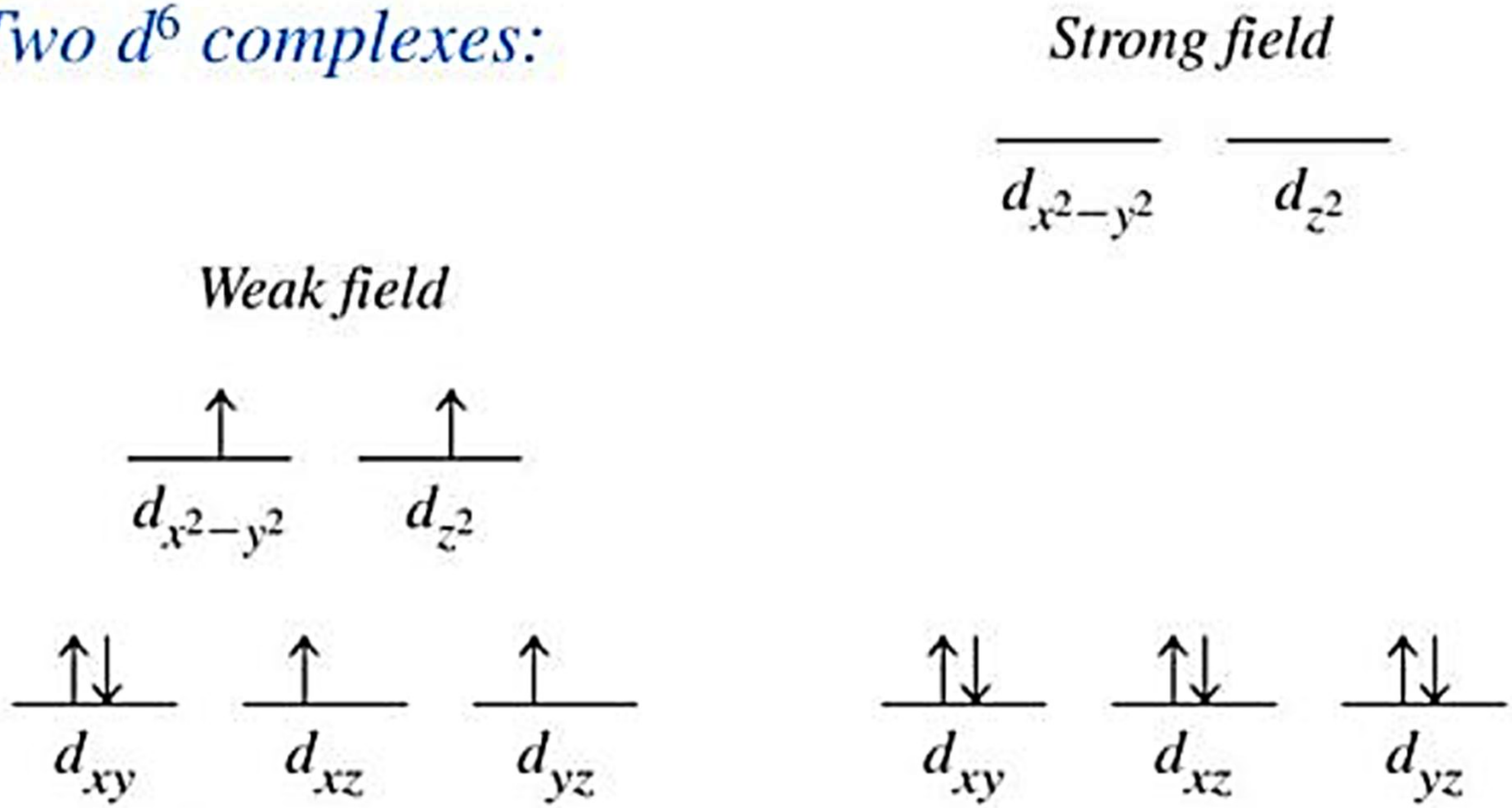
$I^- < Br^- < S^{2-} < \underline{S}CN^- < Cl^- < N_3^-, F^- <$ Urea, $OH^- <$ ox, $O^{2-} < \underline{O}NO^- < H_2O <$

$\underline{N}CS^- <$ py, $NH_3 <$ en $<$ bpy, phen $< \underline{N}O_2^- < CH_3^-, C_6H_5^- < CN^- < CO$

Large Δ
Strong field ligands

Weak Field and Strong Field Ligands

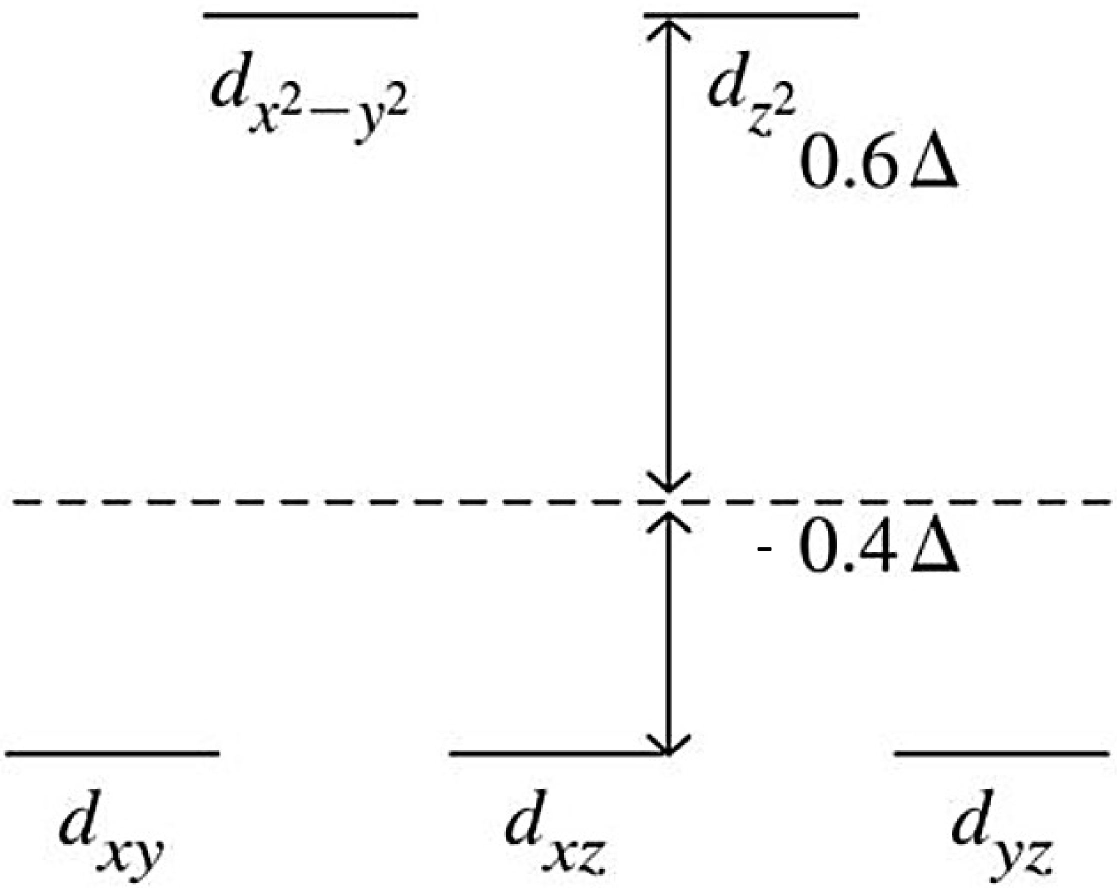
Two d^6 complexes:



High-spin complex
$[CoF_6]^{3-}$
$\Delta < P$

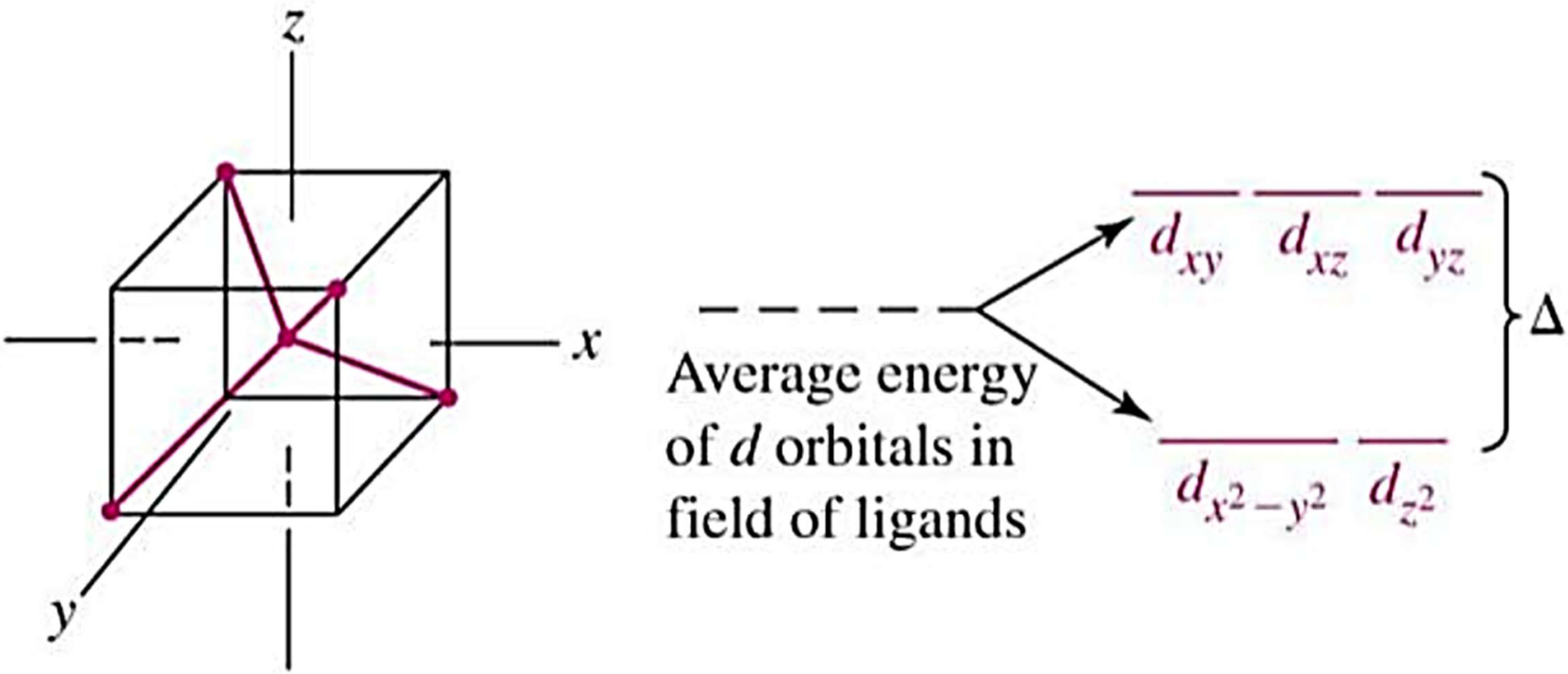
Low-spin complex
$[Co(NH_3)_6]^{3+}$
$\Delta > P$

Energy Effects



CFSE = $-0.4 \times n(t_{2g}) + 0.6 \times n(e_g)\,\Delta_o$

Tetrahedral Crystal Field



(a) (b)

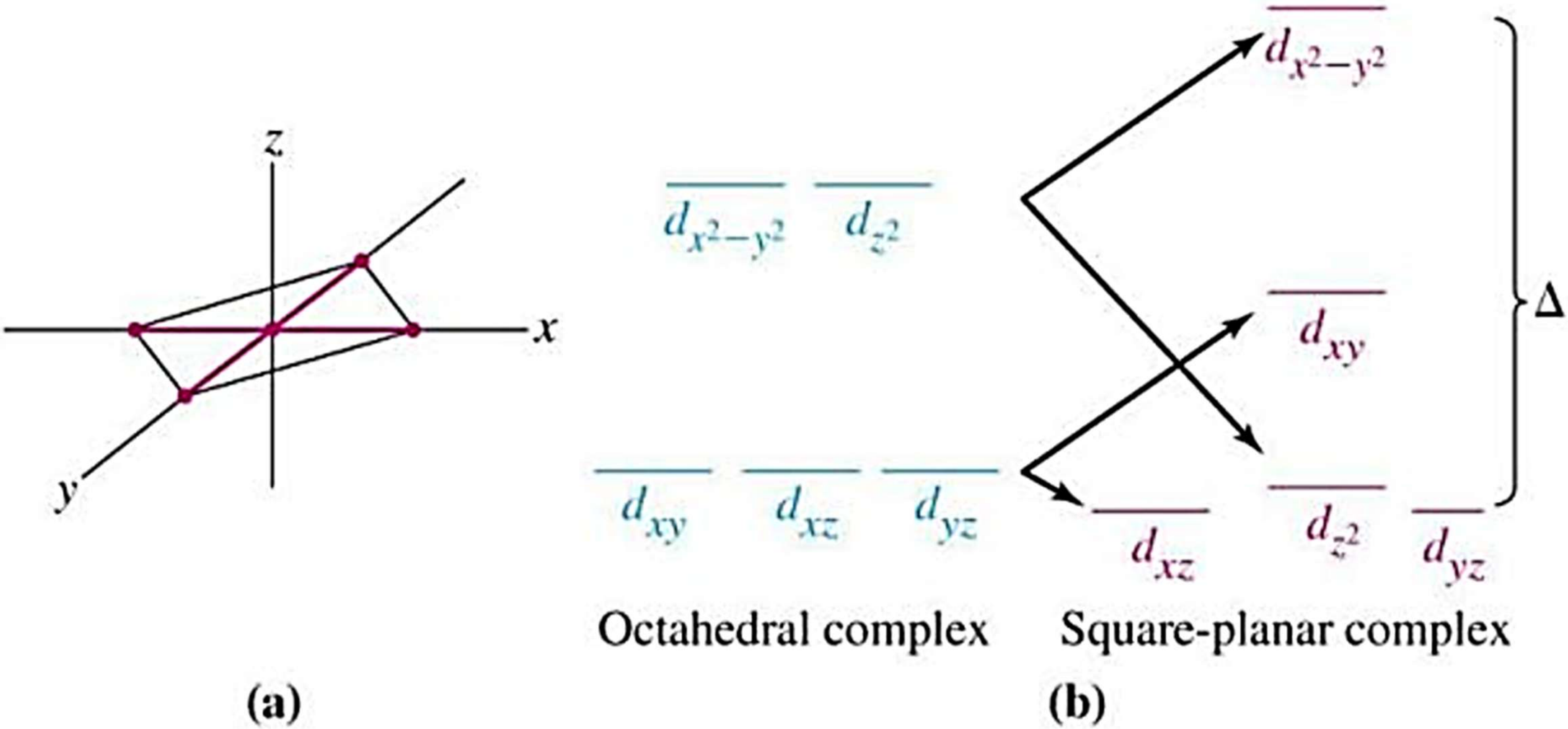
- There are only 4 ligands in the tetrahedral complex, and hence the ligand field is roughly 2/3 of the octahedral field.

- The direction of ligand approach in tetrahedral complex does not coincide with the d-orbitals. This reduces the field by a factor of 2/3. Therefore is roughly 2/3 X 2/3 = 4/9 of Δ_o.

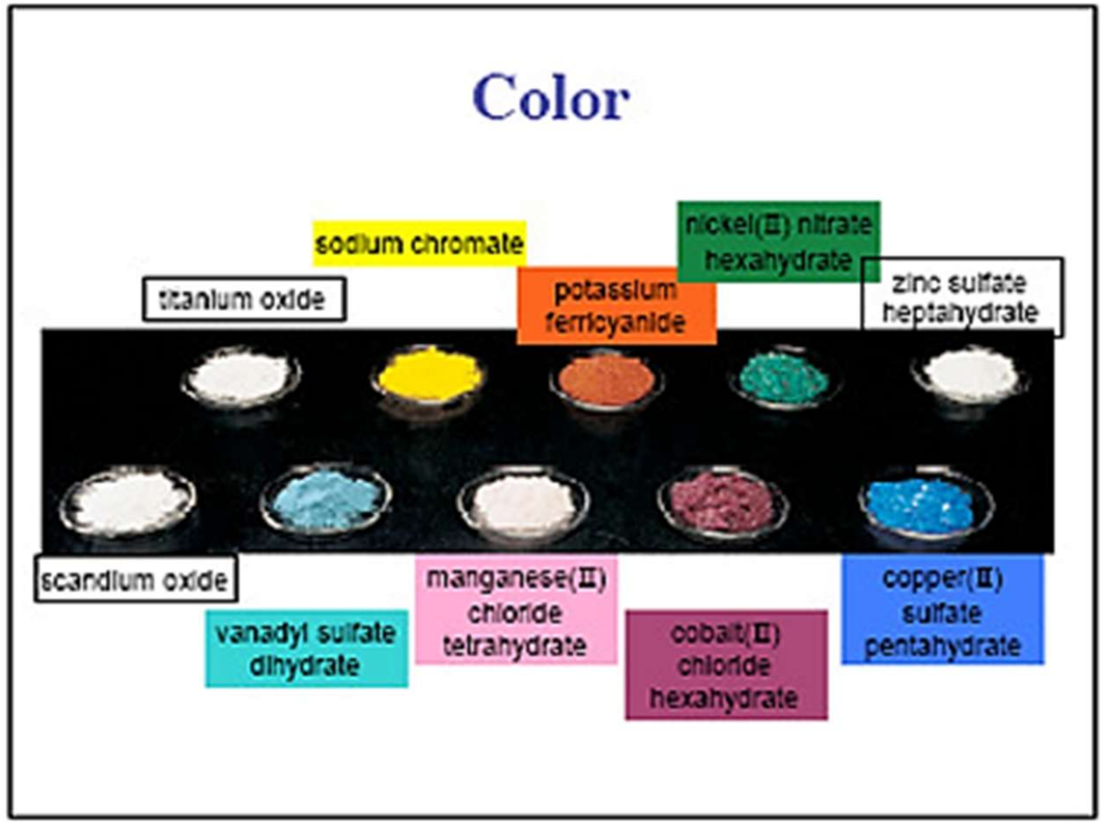
- As a result, all the tetrahedral complexes are high-spin since the CFSE is normally smaller than the pairing energy.

- Hence low spin configurations are rarely observed. Usually, if a very strong field ligand is present, the square planar geometry will be favored.

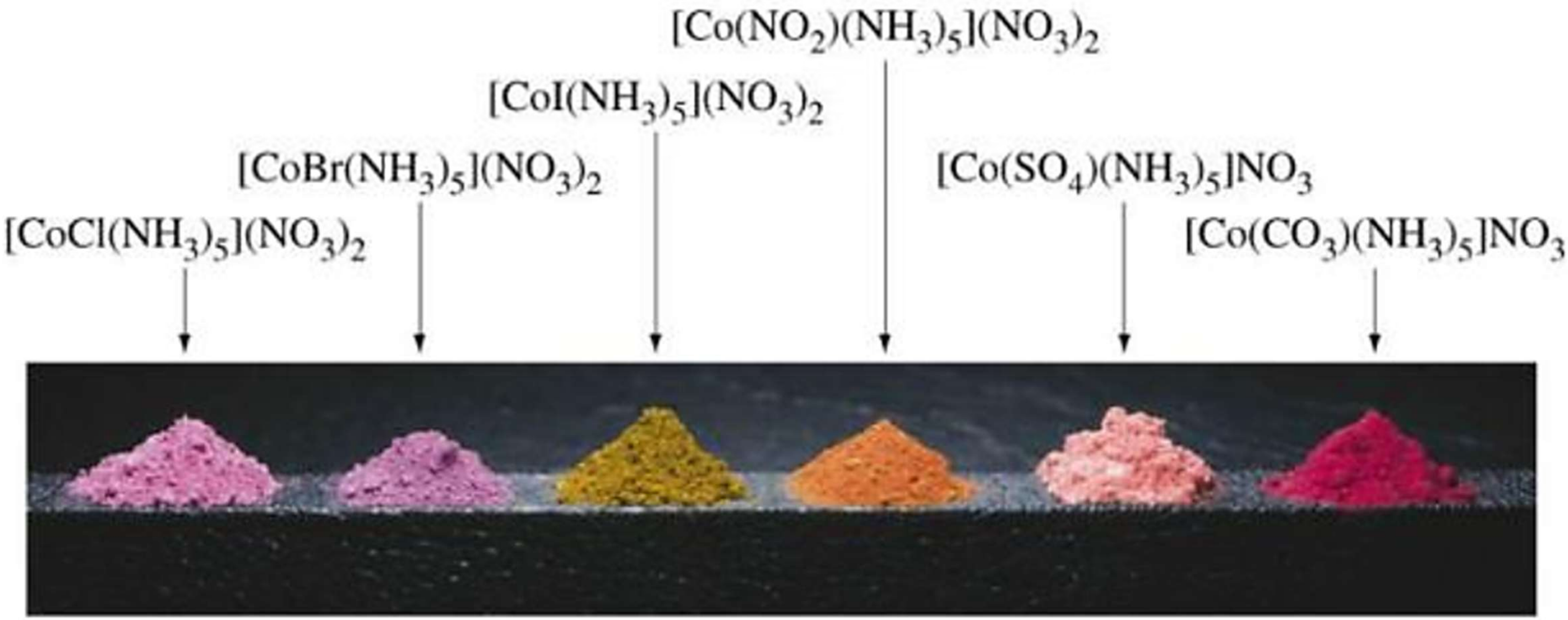
Square Planar Crystal Field



Vanadium metal (center) and in solution as $V^{2+}(aq)$, $V^{3+}(aq)$, $VO^{2+}(aq)$, and $VO_2^{+}(aq)$, (left to right).



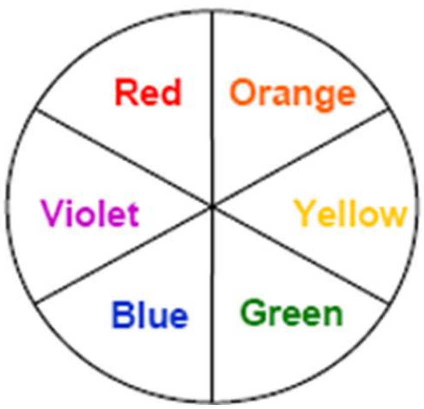
Aqueous solutions of $[Fe(H_2O)_6]^{3+}$ are red, $[Co(H_2O)_6]^{2+}$ are pink, $[Ni(H_2O)_6]^{2+}$ are green, $[Cu(H_2O)_6]^{2+}$ are blue and $[Zn(H_2O)_6]^{2+}$ are colorless. Although the octahedral $[Co(H_2O)_6]^{2+}$ are pink, those of tetrahedral $[CoCl_4]^{2-}$ are blue. The green color of $[Ni(H_2O)_6]^{2+}$ turns blue when ammonia is added to give $[Ni(NH_3)_6]^{2+}$. Many of these facts can be rationalized from CFT.



$[CoCl(NH_3)_5](NO_3)_2$
$[CoBr(NH_3)_5](NO_3)_2$
$[CoI(NH_3)_5](NO_3)_2$
$[Co(NO_2)(NH_3)_5](NO_3)_2$
$[Co(SO_4)(NH_3)_5]NO_3$
$[Co(CO_3)(NH_3)_5]NO_3$

Why we see Color?

When a sample absorbs light, what we see is the sum of the remaining colors that strikes our eyes. If a sample absorbs all wavelength of visible light, none reaches our eyes from that sample, and then the sample appears black. If the sample absorbs no visible light, it is white or colorless. When the sample absorbs a photon of visible light, it is its complementary color we actually see.



The color wheel

Colour of light absorbed	Approx. λ ranges / nm	Colour of light transmitted
Red	700-620	Green
Orange	620-580	Blue
Yellow	580-560	Violet
Green	560-490	Red
Blue	490-430	Orange
Violet	430-380	Yellow

- In general, strong-field ligands cause a large split in the energies of *d* orbitals of the central metal atom (large Δ_{oct}). Transition metal coordination compounds with these ligands are yellow, orange, or red because they absorb higher-energy violet or blue light.

- On the other hand, coordination compounds of transition metals with weak-field ligands are often blue-green, blue, or indigo because they absorb lower-energy yellow, orange, or red light.

Compounds	Δ_o	Color absorbed	Color transmitted
$[Co(H_2O)_6]^{3+}$	Small	Orange	Blue
$[Co(NH_3)_6]^{3+}$	Intermediate	Blue	Orange
$[Co(CN)_6]^{3-}$	Large	Violet	Yellow

$CuSO_4$

$CuSO_4.5H_2O$

- The color of a complex depends on the magnitude of Δ_o, which depends on the structure of the complex.
- For example, the complex $[Cr(NH_3)_6]^{3+}$ has strong-field ligands and a relatively large Δ_o. Consequently, it absorbs relatively high-energy photons, corresponding to blue-violet light, which gives it a yellow color.
- A related complex with weak-field ligands, the $[Cr(H_2O)_6]^{3+}$ ion, absorbs lower-energy photons corresponding to the yellow-green portion of the visible spectrum, giving it a deep violet color.
- The iron(II) complex $[Fe(H_2O)_6]SO_4$ appears blue-green because the high-spin complex absorbs photons in the red wavelengths.
- In contrast, the low-spin iron(II) complex $K_4[Fe(CN)_6]$ appears pale yellow because it absorbs higher-energy violet photons.

$[Fe(H_2O)_6]SO_4$

$K_4[Fe(CN)_6]$

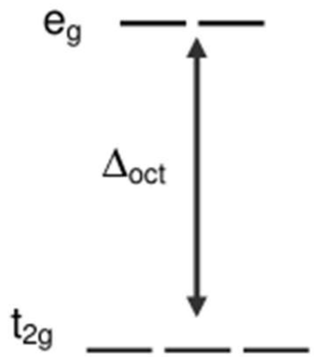
- A ruby is a pink to blood-red colored gemstone that consist of trace amounts of chromium in the mineral corundum Al_2O_3.
- In contrast, emeralds are colored green by trace amounts of chromium within a $Be_3Al_2Si_6O_{18}$ matrix.
- Although the chemical identity of the six ligands is the same in both cases, the Cr–O distances are different because the compositions of the host lattices are different.

- In ruby, the Cr–O distances are relatively short because of the constraints of the host lattice, which increases the d orbital–ligand interactions and makes Δ_o relatively large. Consequently, rubies absorb green light and the transmitted or reflected light is red, which gives the gem its characteristic color.
- In emerald, the Cr–O distances are longer due to relatively large $[Si_6O_{18}]^{12-}$ silicate rings; this results in decreased d orbital–ligand interactions and a smaller Δ_o. Consequently, emeralds absorb light of a longer wavelength (red), which gives the gem its characteristic green color.

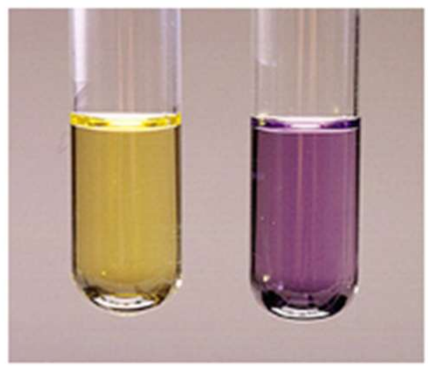
The environment of the transition-metal ion, which is determined by the host lattice, dramatically affects the spectroscopic properties of a metal ion.

$[V(H_2O)_6]^{3+}$
$V(III) = d^2$ ion

violet light absorbed
complex appears yellow



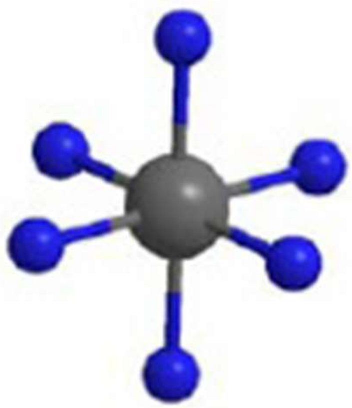
Δ large



$[V(H_2O)_6]^{2+}$
$V(II) = d^3$ ion

yellow light absorbed
complex appears violet

Δ small

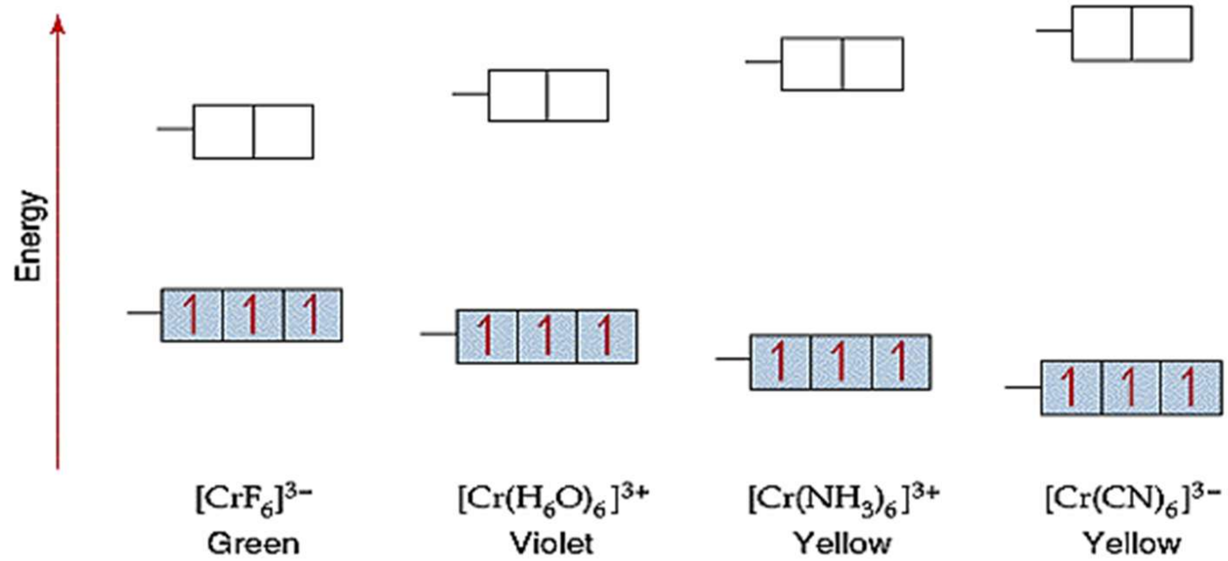


$[Cr(NH_3)_6]^{3+}$

Strong Δ_{oct}
High energy,
violet light absorbed
Complex: yellow

$[Cr(NH_3)_5Cl]^{2+}$

Weak Δ_{oct}
Low energy,
yellow light absorbed
Complex: magenta

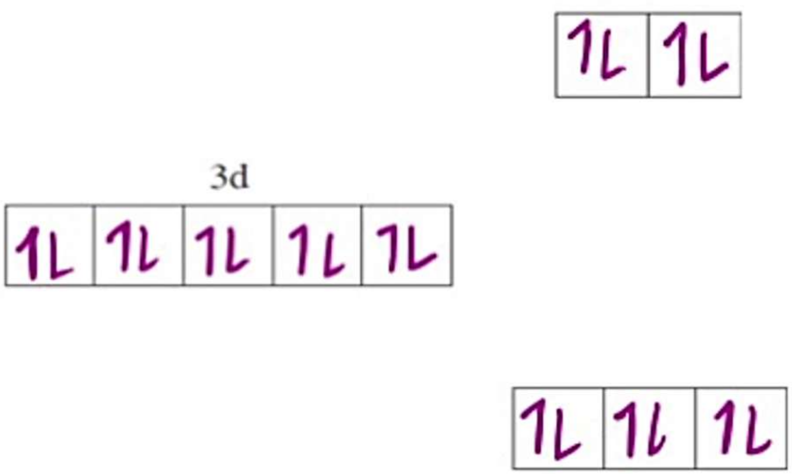


Complexes that contain metal ions of d^{10} electron configuration are usually colourless. Examples are $[Cu(PPh_3)_4]^+$ and $[Zn(H_2O)_6]^{2+}$. One would expect a metal complex with no d-electron to be colourless as well. However, a few of such complexes are strongly coloured, for example, MnO_4^- or $[Cr_2O_7]^{2-}$. The origin of the colour in these complexes is not the d-d transitions, rather due to 'charge transfer'

Colors in Transition Metal Ions & Complexes

- To summarise, the colour of these complexes, is caused by d-d electron transitions. The amount of energy between the lower and higher energy d orbitals, determines the colour of the complex. As this splitting energy (ΔE) varies, so does the frequency of light absorbed, which leads to different colours for different complexes.

- Some transition metal complexes are however colourless. This is down to them having either empty or full d shells so that d-d electron transitions cannot take place. For example, in copper (i) complexes, Cu^{+} has a full 3d shell ($3d^{10}$) and therefore d-d electron transitions cannot take place.



- Similarly, scandium forms colourless complexes as it can only form Sc^{3+} ions with an empty 3d shell. Therefore, once again no d-d electron transitions can take place.

- Remember that zinc is not classed as a transition metal as it doesn't have an atom or a stable ion with a partially filled d shell. It also cannot have d-d electron transitions ($Zn^{2+} = 3d^{10}$). Zinc therefore does not have coloured compounds.

Intensity of Color

How intensity of color is expressed

$$A = \log_{10} \frac{I_o}{I} = \varepsilon\, l\, c$$

The **molar absorption coefficient**, **molar extinction coefficient**, or **molar absorptivity** (ε), is a measurement of how strongly a chemical species absorbs light at a given wavelength. It is an intrinsic property of the species; the actual absorbance A, of a sample is dependent on the pathlength, ℓ, and the concentration, c, of the species via the **Beer–Lambert law,**

Selection rules:

Electronic transitions in a complex are governed by Selection rules

A selection rule is a quantum mechanical rule that describes the types of quantum mechanical transitions that are permitted.

They reflect the restrictions imposed on the state changes for an atom or molecule during an electronic transition.

Transitions not permitted by selection rules are said **forbidden**, which means that theoretically they must not occur (but in practice may occur with very low probabilities).

Laporte Selection Rule

Otto Laporte
German American Physicist

Statement : Only allowed transitions are those occurring with a change in parity (flip in the sign of *one* spatial coordinate.) OR During an electronic transition the azimuthal quantum number can change only by ± 1 ($\Delta l = \pm 1$)
The Laporte selection rule reflects the fact that for light to interact with a molecule and be absorbed, there should be a change in dipole moment.

Practical meaning of the Laporte rule

Allowed transitions are those which occur between gerade to ungerade or ungerade to gerade orbitals

Allowed

$g \rightarrow u$ & $u \rightarrow g$

Not allowed (FORBIDDEN)

$g \nrightarrow g$ & $u \nrightarrow u$

$t_{2g} \rightarrow e_g$ is forbidden OR
According to Laporte selection rule
d→d transitions are not allowed !

Gerade = symmetric w r t centre of inversion
Ungerade = antisymmetric w r t centre of inversion

This rule affects Octahedral and Square planar complexes as they have center of symmetry.

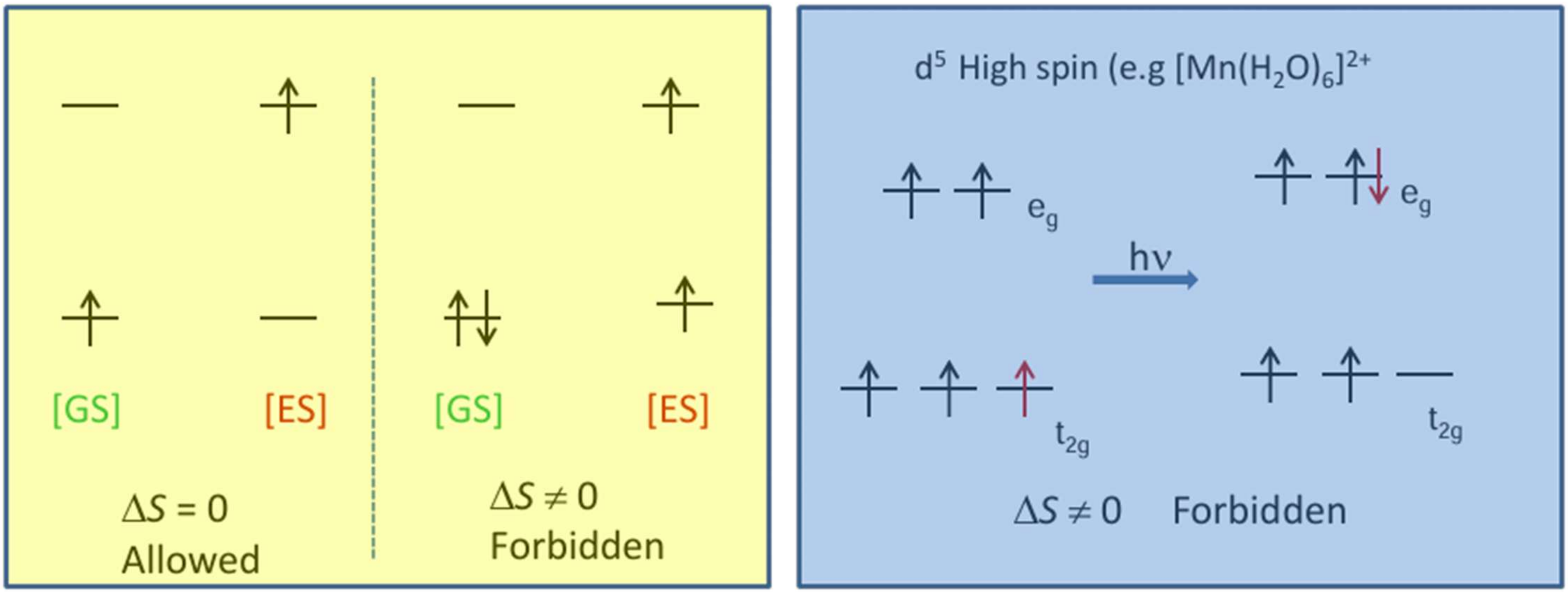
Tetrahedral complexes do not have center of symmetry: therefore this rule does not apply

Spin Selection Rule

Statement : This rule states that transitions that involve a change in spin multiplicity are forbidden. According to this rule, any transition for which $\Delta S = 0$ is allowed and $\Delta S \neq 0$ is forbidden

Practical significance of the Spin Selection rule

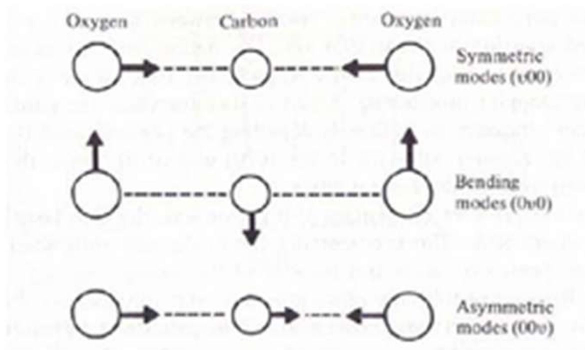
During an electronic transition, the electron should not change its spin



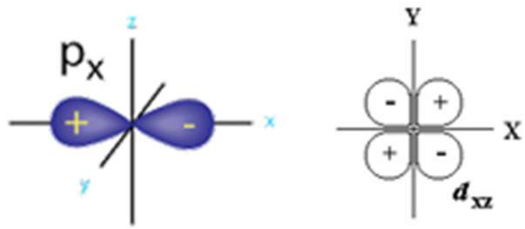
Why do we see 'forbidden' transitions at all

There are three mechanisms that allow 'forbidden' electronic transitions to become somewhat 'allowed' resulting in some intensity of the color expected.

1) **Vibronic Coupling**: During some unsymmetrical vibrations of a molecule there can be a temporary/transient loss of the centre of symmetry. Loss of center of symmetry helps to overcome the Laporte selection rule. Also time required for an electronic transition to occur (lifetime 10^{-18} sec) is much less than the time required for a vibration to occur (lifetime 10^{-13} sec

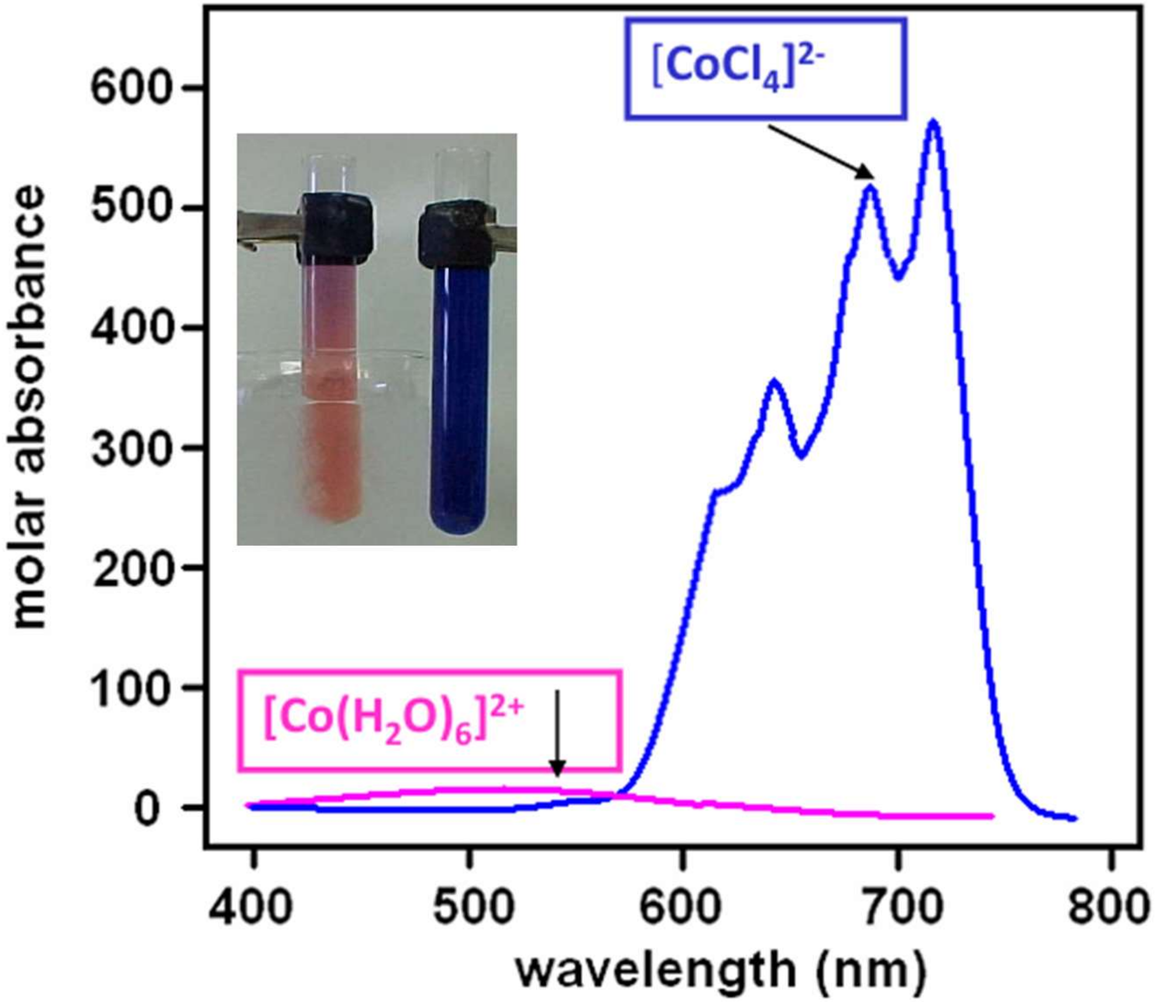


2) **Mixing of states**: The states in a complex are never pure, and so some of the symmetry properties (g or u) of neighboring states become mixed into those of the states involved in a 'forbidden' transition. For example mixing of d (gerade) and p (ungerade) orbitals results in partial breakdown of the Laporte rule

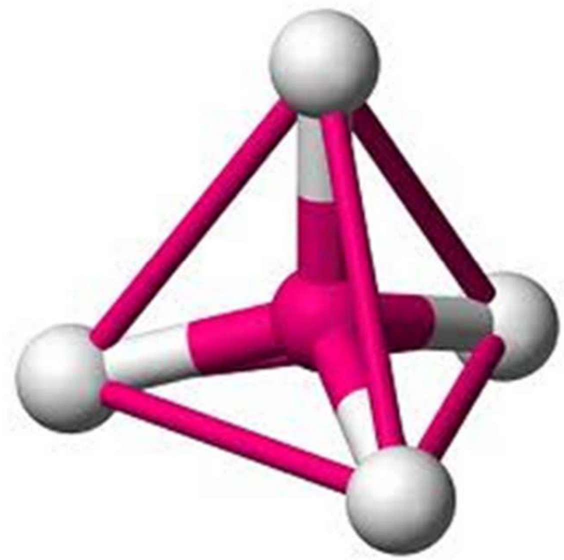


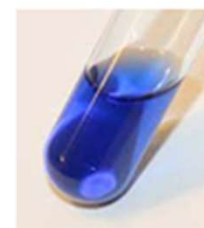
3) **Spin orbit coupling:** Partial lifting of the spin selection rule is possible when there is coupling of the spin and orbital angular momentum, known as the spin-orbit coupling (common in heavier transition metals)

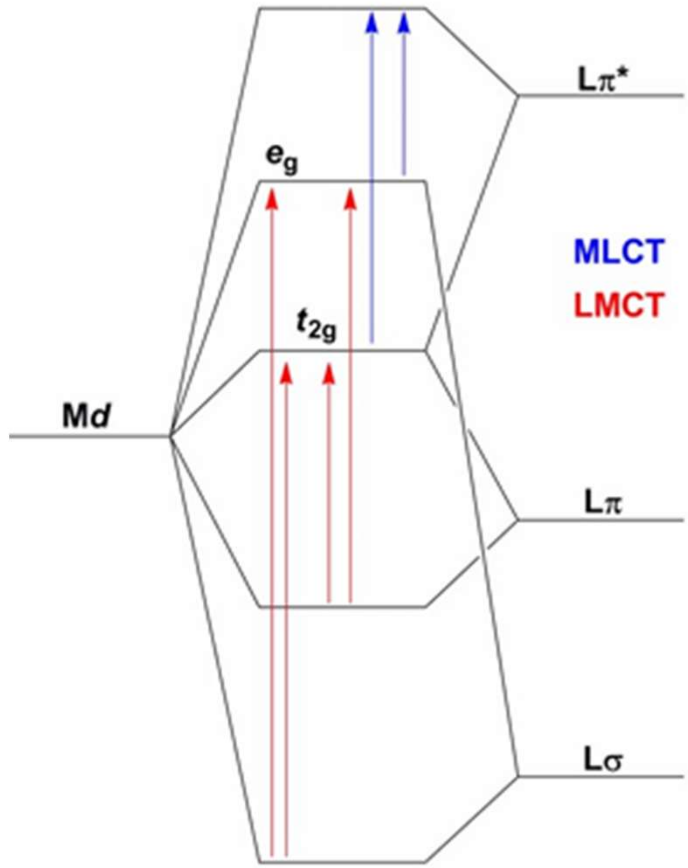
$[Mn(H_2O)_6]^{2+}$



Intense d-d bands in the blue tetrahedral complex [CoCl$_4$]$^{2-}$, as compared with the much weaker band in the pink octahedral complex [Co(H$_2$O)$_6$]$^{2+}$. This difference arises because the T_d complex has no center of symmetry, helping to overcome the g→g Laporte selection rule.



Transition type	Example	Typical values of ε /$m^2 mol^{-1}$	
Spin forbidden, **Laporte forbidden** **(partly allowed by spin–orbit coupling)**	$[Mn(H_2O)_6]^{2+}$	0.1	
Spin allowed (octahedral complex), **Laporte forbidden** **(partly allowed by vibronic coupling and d-p mixing)**	$[Co(H_2O)_6]^{2+}$	1 - 10	
Spin allowed (tetrahedral complex), **Laporte allowed** (but still retain some original character)	$[CoCl_4]^{2-}$	50 - 150	 
Spin allowed, **Laporte allowed** **e.g. charge transfer bands**	$KMnO_4$	1000 - 10^6	



$KMnO_4$

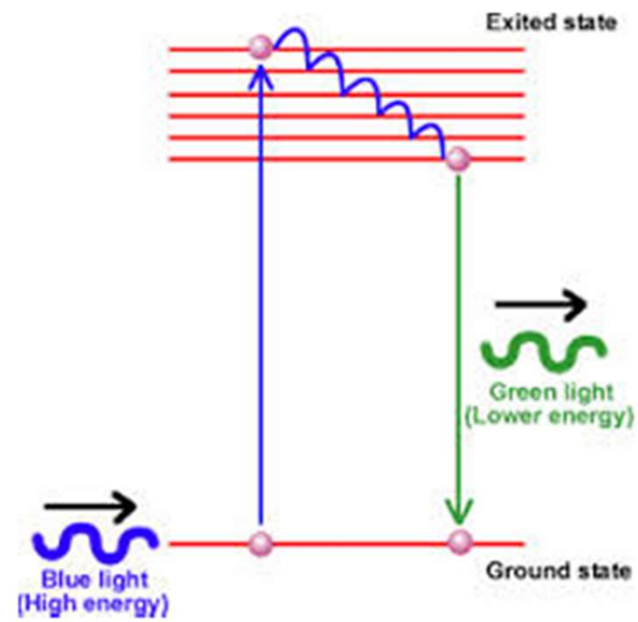
Prussian Blue

Ferric Thiocyanate

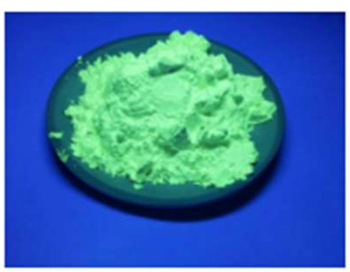
Pot. Dichromate

$Cr(CO)_6$
$Mo(CO)_6$ } Colorless
$W(CO)_6$

However some complexes also show a phenomenon known as Fluorescence



Fluorescence is the emission of light by a substance that has absorbed light or other electromagnetic radiation. In most cases the emitted light has a longer wavelength and a lower energy than the absorbed radiation.
So if such complexes are irradiated with UV light, the excited electron will lose some energy and then fall back to ground state emitting (fluorescent) light in the visible range



Zn_2SiO_4

Advantages and Disadvantages of Crystal Field Theory

Advantages over Valence Bond theory

1. *Explains colors of complexes*
2. *Explains magnetic properties of complexes (without knowing hybridization) and temperature dependence of magnetic moments.*
3. *Classifies ligands as weak and strong*
4. *Explains anomalies in physical properties of metal complexes*
5. *Explains distortion in shape observed for some metal complexes*

Disadvantages or drawbacks

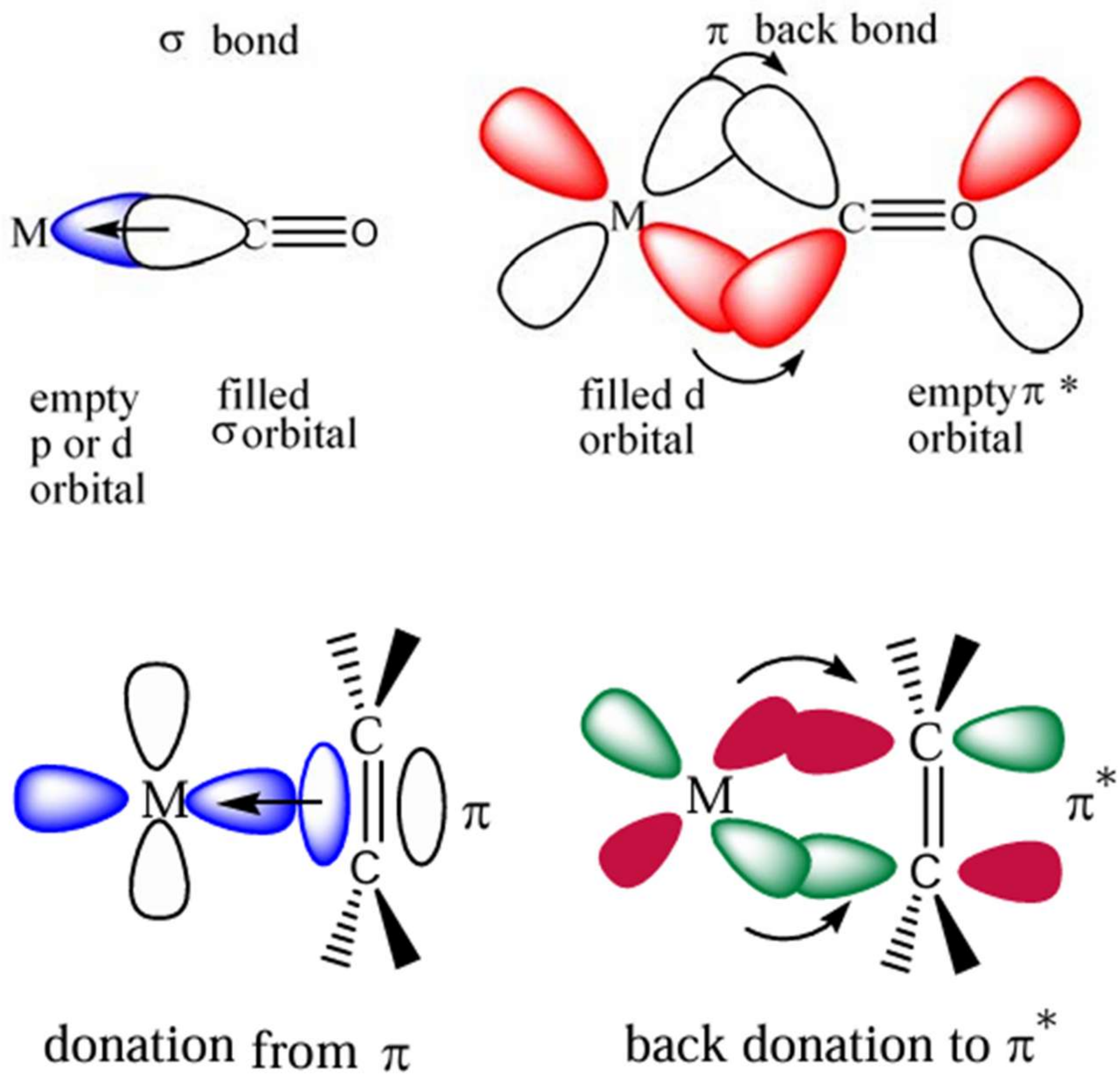
1. *Evidences for the presence of covalent bonding (orbital overlap) in metal complexes have been disregarded.*
   e.g Does not explain why CO although neutral is a very strong ligand

2. *Cannot predict shape of complexes (since not based on hybridization)*
3. *Charge Transfer spectra not explained by CFT alone*

Organometallic Chemistry

An area which bridges organic and inorganic chemistry.

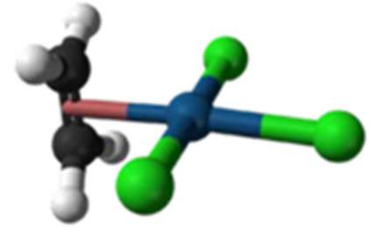
The complex has one or more metal-carbon Bonds.



Zeise's Salt: The first transition metal Organometallic Compound

W C Zeise, Danish pharmacist, **1789- 1847**

$$K_2PtCl_4 + C_2H_5OH \longrightarrow K[(C_2H_4)PtCl_3].H_2O + KCl$$



Discovery 1827

Structure ~ 150 years later

Also father of the chemistry of mercaptans R-SH

'The breakthrough, the isolation of a pure, crystalline compound came when Zeise added potassium chloride to a concentrated $PtCl_4$ /ethyl alcohol reaction solution and evaporated the resulting solution. Beautiful lemon yellow crystals, often one half inch or more in length were isolated. On longer exposure to air and light, they gradually became covered with a black crust. They contained water of hydration, which was lost when they were kept over concentrated sulfuric acid in vacuo or when heated to around 100°C. Chemists in those days often reported how the compounds that they had prepared tasted. Zeise described the taste of this potassium salt as metallic, astringent and long lasting.'

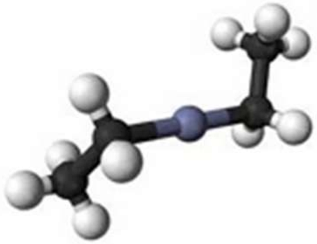
Dietmar Seyferth, Organometallics, 2001, 20, 2

First σ bonded Organometallic Compound- Diethyl zinc

Edward Frankland
1825-1899

Frankland coined the term "Organometallic"

$$3\ C_2H_5I + 3\ Zn \rightarrow (C_2H_5)_2Zn + C_2H_5ZnI + ZnI_2$$



Student of Robert Bunsen (Bunsen burner fame!). Prepared diethyl zinc while trying to make ethyl radicals.

As the early 1850s English chemist Edward Frankland described flasks exploding, throwing bright green flames across his lab, as he heroically distilled dialkylzinc compounds under an atmosphere of hydrogen.

Metal carbonyls

Ludwig Mond 1839-1909

Father of Metal Carbonyl Chemistry

Founder of Imperial Chemical Industry, England

Metal carbonyls

The Mond process of Nickel purification

$NiO + H_2$ (from Syn gas) $\xrightarrow{200\ ^\circ C}$ Impure Ni (Fe and Co) $+ H_2O$

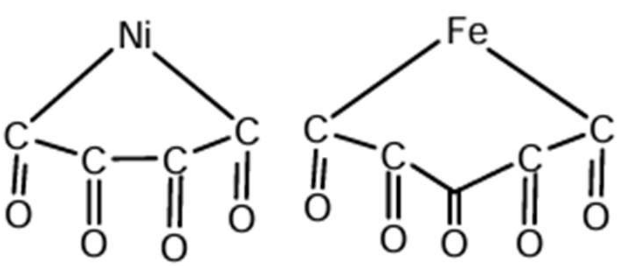
Impure Ni $\xrightarrow{\text{excess CO, } 50-60\ ^\circ C}$ $Ni(CO)_4$ (g) bp 42 °C

$Ni(CO)_4$ (g) $\xrightarrow{220-250\ ^\circ C}$ Ni(s) + 4 CO

$Ni(CO)_4$, $Fe(CO)_5$, $Co_2(CO)_8$, $Mo(CO)_6$

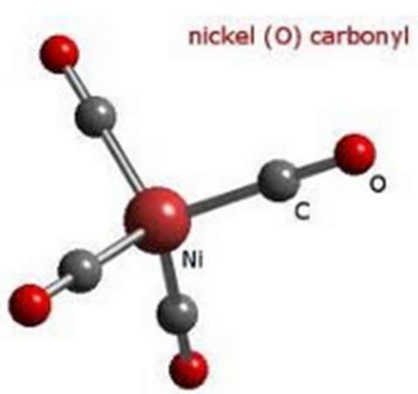
1890 1891 1910

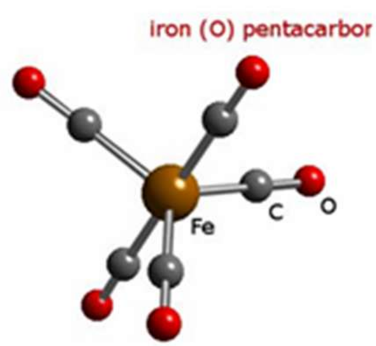
'*Mond nickel company*' was making over 3000 tons of nickel in 1910 with a purity level of 99.9%



$Ni(CO)_4$ $Fe(CO)_5$

1890-1930 textbooks



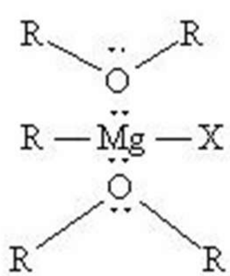


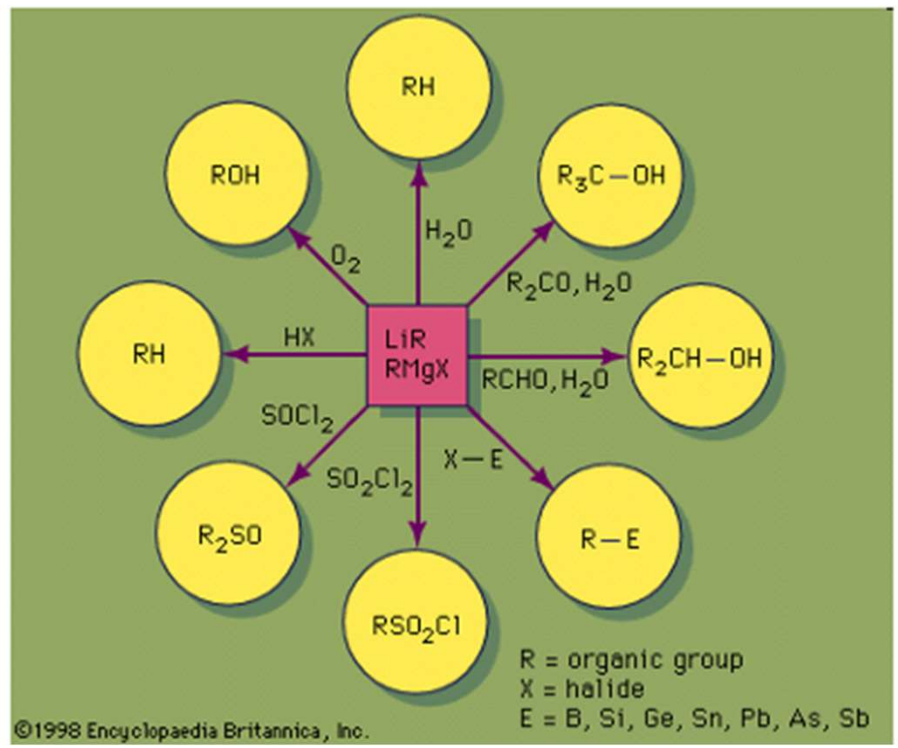
$Mo(CO)_6$

The Grignard Reagent

François Auguste Victor Grignard 1871-1935

He was the student of Philippe Barbier (Barbier reaction [Zn]) He discovered the Grignard reaction [Mg]) in 1900. He became a professor at the University of Nancy in 1910 and was awarded the Nobel Prize in Chemistry in 1912.





Ferrocene: Pathbreaking discovery of a sandwich compound

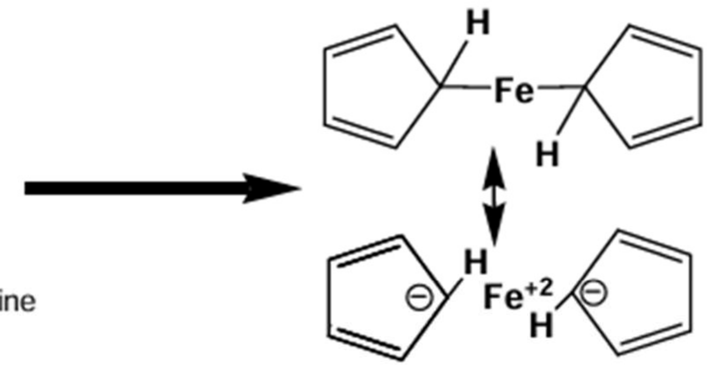
Pauson

$FeCl_3$ + CpMgBr

Kealy and Pauson

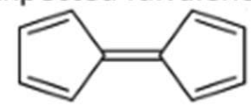
Fe + Cp

Miller, Tebboth and Tremaine



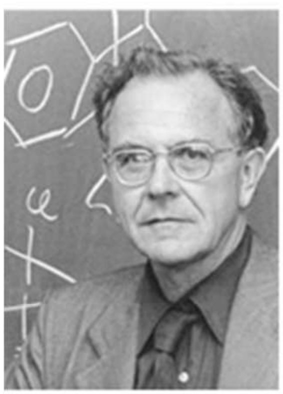
Kealy

expected fulvalene

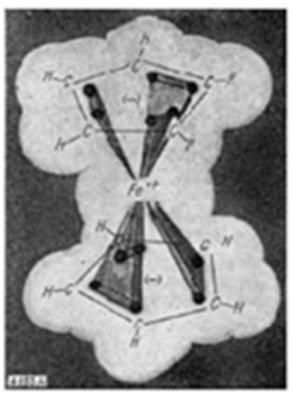


A new type of organo-iron compound, *Nature 1951*

Dicyclopentadienyl iron, J. Chem. Soc., 1952



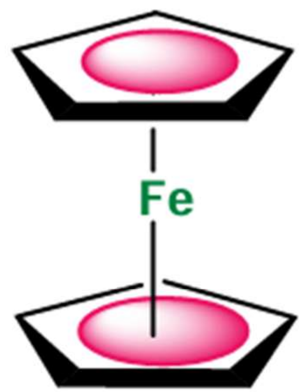
G. Wilkinson

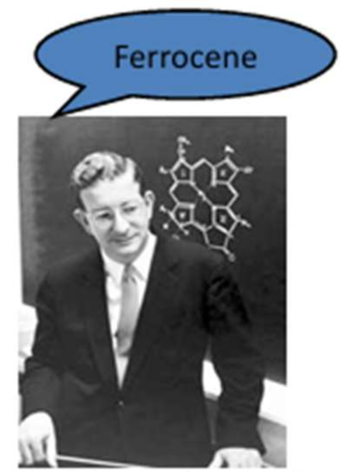


1973 Nobel Prize

'sandwich compounds'

E. O. Fischer





R. B. Woodward

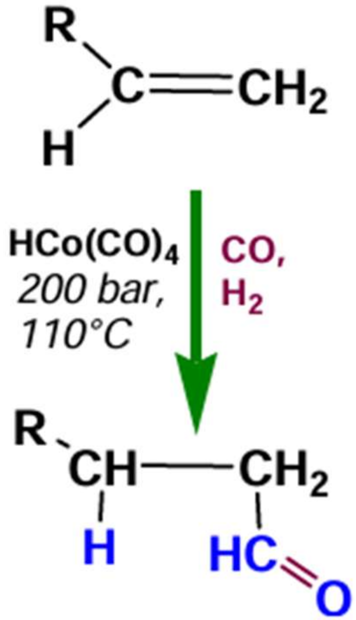
1965 Nobel Prize

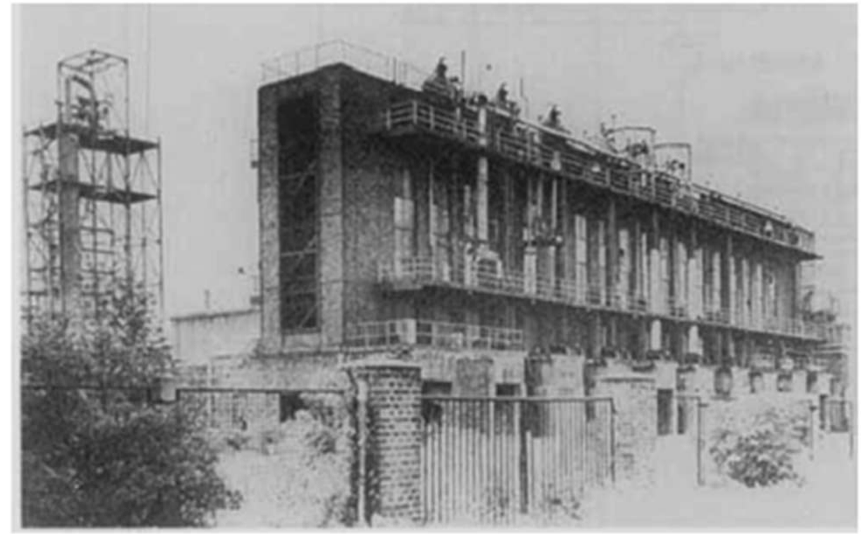
'art of organic synthesis'

Wilkinson, Rosenblum, Whitney, Woodward, *J. Am. Chem. Soc.*, 1952

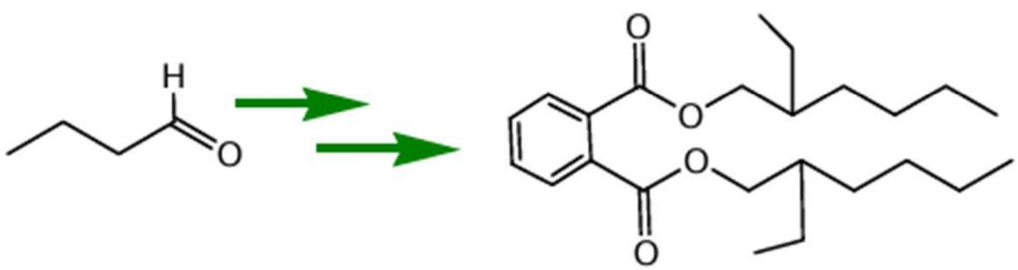
First organometallics in homogeneous catalysis: The Hydroformylation (1938)

Otto Roelen
Pioneer in Industrial homogeneous catalysis
(1897-1993)



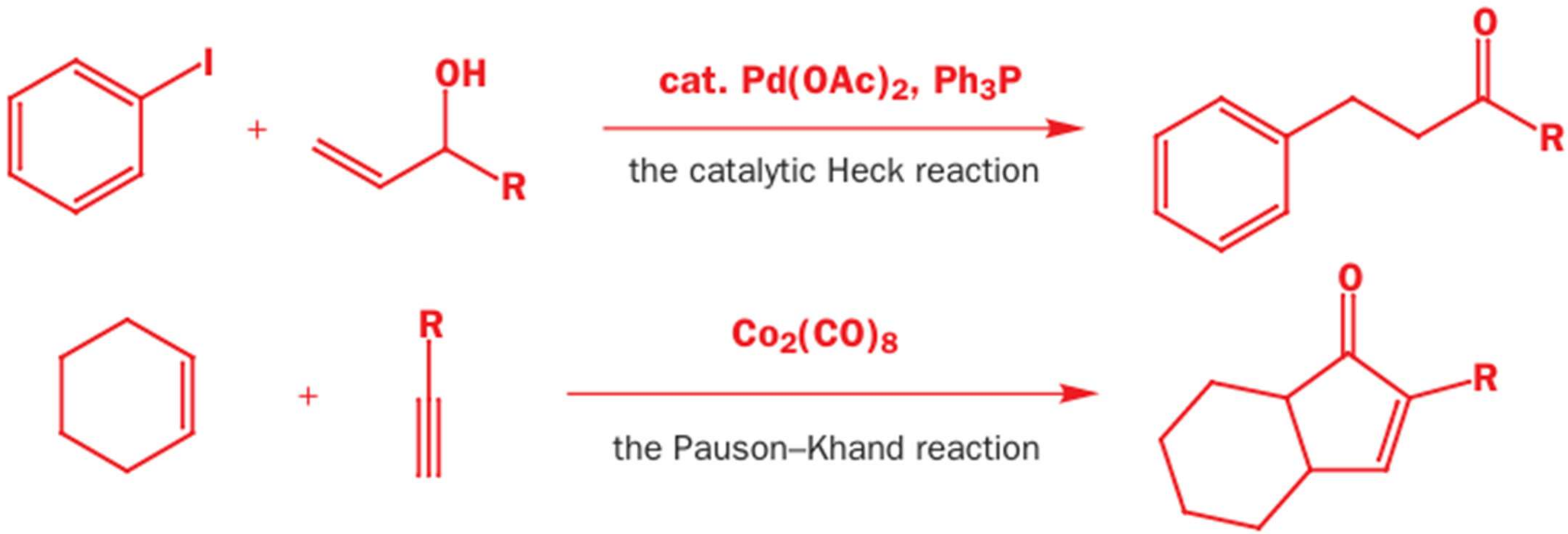


First Industrial plant- hydroformylation



diethylhexylphthalate [DEHP]
Plasticizer

detergents


I
OH
R
cat. Pd(OAc)2, Ph3P
the catalytic Heck reaction
O
R
R
Co2(CO)8
the Pauson–Khand reaction
O
R

- An ideal catalyst is a complex that is stable in the resting state, for storage, but quickly becomes activated in solution, perhaps by loss of a ligand, allowing inter action with the substrate.
- If a complex satisfies the 18-electron rule for a stable metal complex it means that the metal at the centre of the complex has the noble gas configuration of 18 electrons in the valence shells.
- The total of 18 is achieved by combining the electrons that the metal already possesses with those donated by the coordinating ligands.

Group	**IVB (4)**	**VB (5)**	**VIB (6)**	**VIIB (7)**	**VIIIB (8, 9, and 10)**			**1A (11)**
Number of valence electrons	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**
3d	**Ti**	V	**Cr**	**Mn**	**Fe**	**Co**	**Ni**	**Cu**
4d	**Zr**	Nb	**Mo**	Tc	**Ru**	**Rh**	**Pd**	Ag
5d	Hf	Ta	**W**	Re	**Os**	Ir	**Pt**	Au

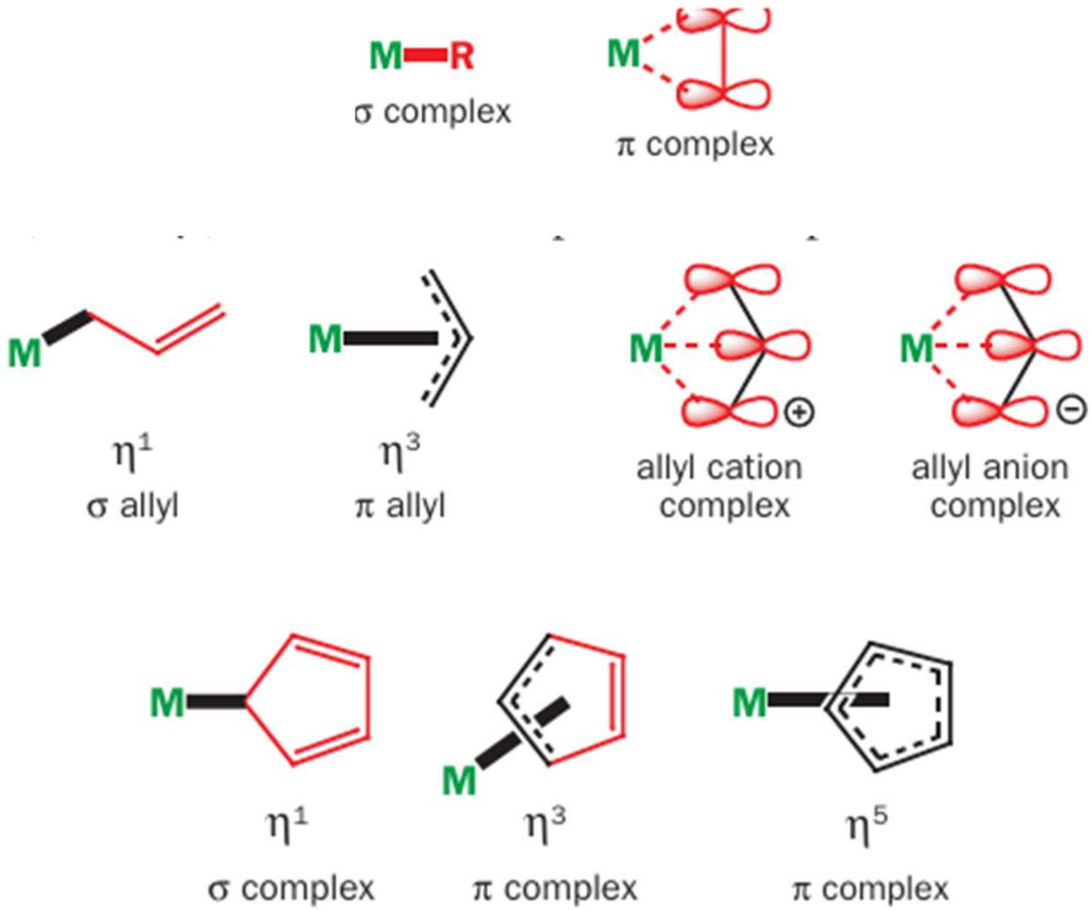
Metals to the left-hand side of this list obviously need many more electrons to make up the magic 18. Chromium, for example, forms stable complexes with a benzene ring, giving it six electrons, and three molecules of carbon monoxide, giving it two each: 6 + 6 + 2 + 2 + 2 = 18. Palladium is happy with just four triphenylphosphines (Ph_3P:) giving it two each: 10 + 2 + 2 + 2 + 2 = 18

However, there are exceptions to the 18-electron rule including complexes of Ti, Zr, Ni, Pd, and Pt, which all form stable 16-electron complexes. The so-called platinum metals Ni, Pd, and Pt are extremely important in catalytic processes. The stable 16-electron configuration results from a high energy vacant orbital caused by the complex adopting a square planar geometry. The benefit of this vacant orbital is that it is a site for other ligands in catalytic reactions

a 16-electron Pd(II) complex

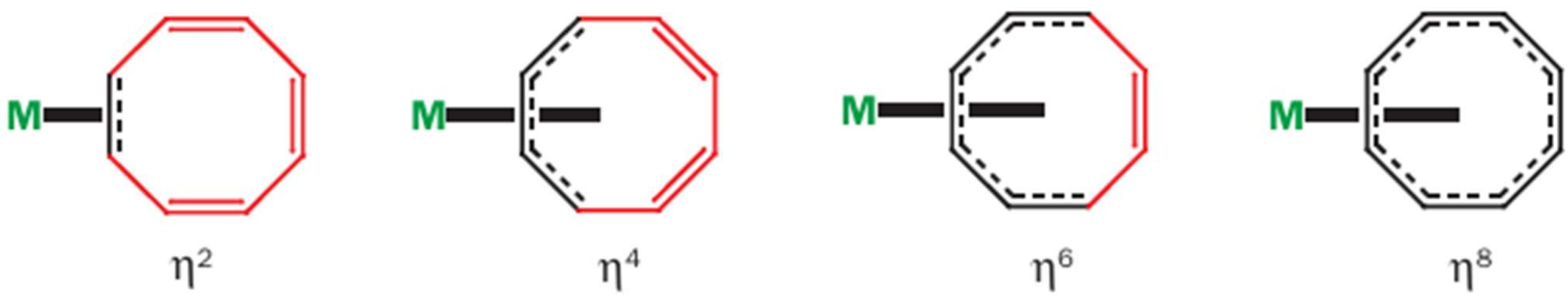
- Transition metals can have a number of ligands attached to them and each ligand can be attached in more than one place.
- This affects the reactivity of the ligand and the metal because each additional point of attachment means the donation of more electrons.
- The number of atoms involved in bonding to the metal is shown by the hapto number η.
- A simple Grignard reagent is η^1 (pronounced ‘eta-one’) as the magnesium is attached only to one carbon atom. A metal–alkene complex is η^2 because both carbon atoms of the alkene are equally involved in bonding to the metal.

η^1 η^2

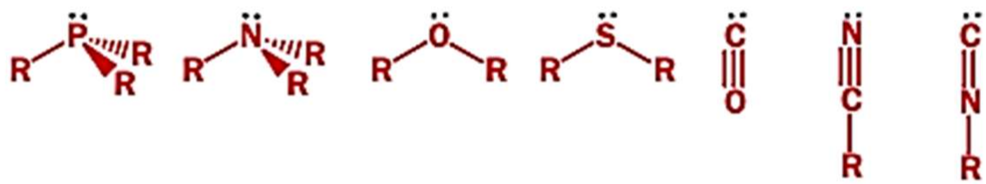


Cyclopentadienyl anion can act as a σ ligand (η^1), an allyl ligand (η^3), or, most usually, as a cyclopentadienyl ligand (η^5). The distinction is very important for electron counting as these three different situations contribute 2, 4, or 6 electrons, respec tively, to the complex.

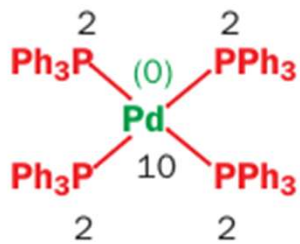
Neutral ligands can also bond in a variety of ways. Cyclooctatetraene can act as an alkene (η^2), a diene (η^4), a triene (η^6), or a tetraene (η^8), and the reactivity of the ligand changes accordingly. These are all π-complexes with the metal above or below the black portion of the ring and with the thick bond to the metal at right angles to the alkene plane.



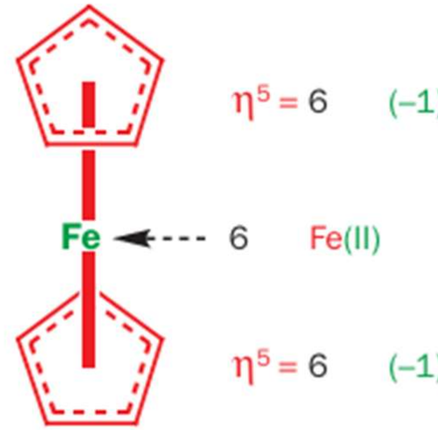
Ligand characteristics

	Formal charge	Electrons donated
anionic ligands		
$Cl^{\ominus}$ $Br^{\ominus}$ $I^{\ominus}$ $^{\ominus}CN$ $^{\ominus}OR$ $^{\ominus}H$ $^{\ominus}$alkyl	–1	2
neutral σ-donor ligands		
	0	2

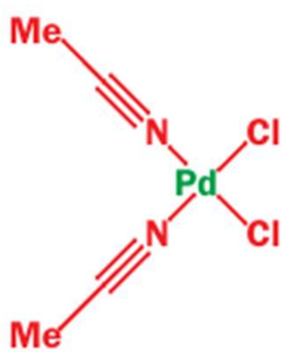
	Hapto number	Formal charge	Electrons donated
unsaturated σ- or π-donor ligands			
aryl, σ-allyl	η^1	–1	2
olefins	η^2	0	2
π-allyl cation	η^3	+1	2
π-allyl anion	η^3	–1	4
diene—conjugated	η^4	0	4
dienyls, cyclopentadienyls (anions)	η^5	–1	6
arenes, trienes	η^6	0	6
trienyls, cycloheptatrienyls (anions)	η^7	–1	8
cyclooctatetraene	η^8	0	8
carbene, nitrene, oxo	η^1	0	2



total: 4 x 2e + 10e = 18e



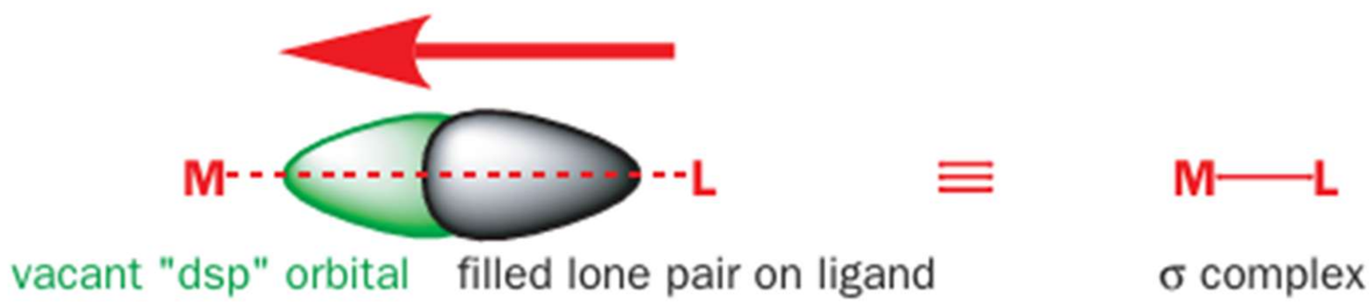
total: 3 x 6e = 18e



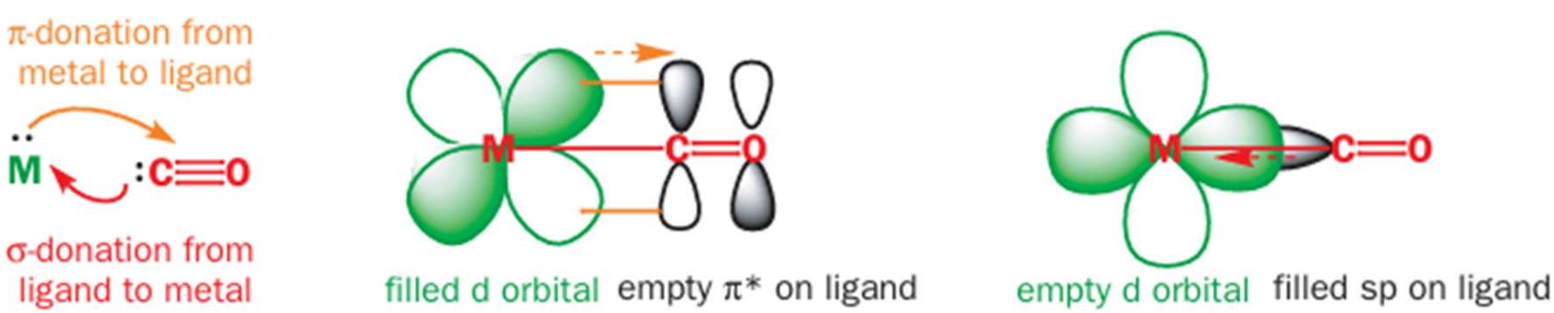
a Pd(II) complex

Transition metal complexes exhibit special bonding

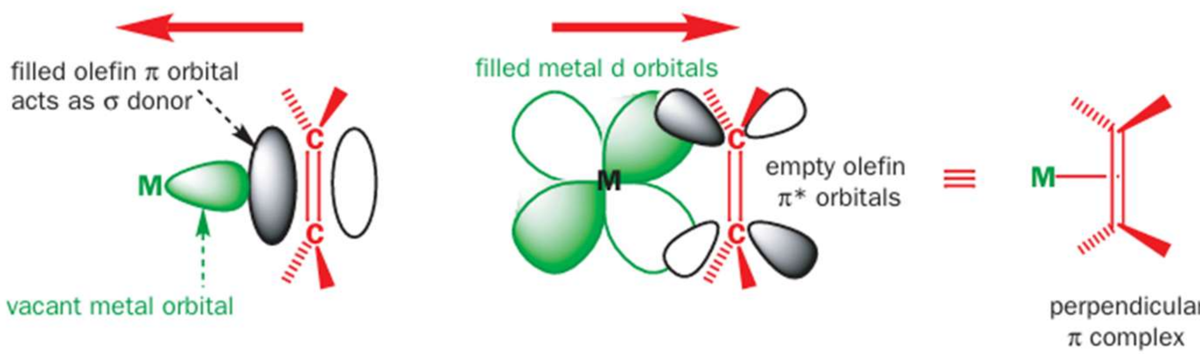
The majority of ligands have a lone pair of electrons in a filled sp^n type orbital that can overlap with a vacant metal 'dsp' orbital, derived from the vacant d, p, and s orbitals of the metal, to form a conventional two-electron two-centre σ bond. Ligands of this type increase the electron density on the central metal atom. This is the sort of bond that used to be called 'dative covalent' and represented by an arrow.



- A bonding interaction is also possible between any filled d orbitals on the metal and vacant ligand orbitals of appropriate symmetry such as π* orbitals. This leads to a reduction of electron density on the metal and is known as back-bonding.
- The ligand (CO) donates the lone pair on carbon into an empty orbital on the metal while the metal donates electrons into the low-energy π* orbital of CO. Direct evidence for this back-bonding is an increase in the C–O bond length and a lowering of the infrared stretching frequency from the population of the π * orbital of the carbonyl.



When an unsaturated ligand such as an alkene approaches the metal sideways to form a π complex, similar interactions lead to bonding. The filled π orbitals of the ligand bond to empty d orbitals of the metal, while filled d orbitals on the metal bond to the empty π * orbitals of the ligand. The result is a π complex with the metal–alkene bond perpendicular to the plane of the alkene. The bond has both s and p character.



Coordination to a metal by any of these bonding methods changes the reactivity of the ligands dramatically